10.08.2026: Kangaroo rat Measurement outliers and initial graphs

Set up

```
library(ggplot2)
library(brms)
```

```
Loading required package: Rcpp
```

```
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':

    ar
```

```
library(ggeffects)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v lubridate 1.9.4     v tibble    3.3.1
v purrr     1.2.0     v tidyr     1.3.2
```

```
-- Conflicts --------------------------------------- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(readxl)
library(here)
```

```
here() starts at C:/Users/Rwncr/Documents/OFFICIALKRATBUSINESS/Krat-Senior-thesis-2026-27
```

```
library(dplyr)
library(ggtext)

#Load in clean data set
df <- read.csv("RC2026_KRAT_DATA.08.01.csv")
head(df)
```

```
  OBJECTID                                GlobalID      Collection_Date
1        1 {12890BA2-1619-41C5-AE55-5B1D7D252FFB} 2025-11-08T16:01:31Z
2        2 {A720CCF9-2B13-4707-B8D2-426CE01002E3} 2025-11-08T16:19:58Z
3        3 {E90F2F02-96F0-4E67-A73E-A4B3C000C67D} 2025-11-08T17:07:49Z
4        4 {865AF038-9B16-44EE-91E3-5A336500AB37} 2025-11-08T17:19:22Z
5        5 {B464F9EF-9309-4CA1-8F9C-1D45FF69CA99} 2025-11-09T16:19:56Z
6        6 {909C48BB-D2E9-444E-BCDA-7D063BB53D50} 2025-11-09T17:49:43Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1          <NA>            51.00       5.00        3.00           39.00
2    RC2025-22             48.38       5.34        3.03           34.76
3    RC2025-23             51.19       5.83        3.51           34.62
4    RC2025-24             52.34       5.74        3.96           37.41
5          <NA>            52.90       5.57        2.74           36.15
6          <NA>               NA         NA          NA              NA
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1            18.00                15.00       51.00            10          7
2            23.80                21.52       45.51             8          7
3            20.38                19.96       47.27            10          6
4            24.03                18.01       47.00             9          7
5            20.51                22.52       47.70            10          8
6               NA                   NA          NA            NA         NA
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         17         15          46         233         122   Non-Reproductive
2         18         15          43         211          93            Unknown
```

```
3         17         17          45         236          NA               <NA>
4         20         15          46         233         110   Non-Reproductive
5         20         16          48         239         128   Non-Reproductive
6         NA         NA          NA          NA          NA               <NA>
      Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal            Species
1  Female      Adult                 82                100 Dipodomys venustus
2 Unknown      Adult                 76                 94 Dipodomys venustus
3    <NA>       <NA>                 NA                 88 Dipodomys venustus
4    Male      Adult                 89                110 Dipodomys venustus
5  Female      Adult                 82                105 Dipodomys venustus
6    <NA>       <NA>                 NA                 NA Dipodomys venustus
  Genus  Tag YN_Recapture Location                 Date Latitude Longitude
1    Di <NA>           No Hastings 2025-11-08T16:01:37Z       NA        NA
2    Di  108          Yes Hastings 2025-11-08T16:20:10Z       NA        NA
3    Di  948          Yes Hastings 2025-11-08T17:09:58Z       NA        NA
4    Di  917           No Hastings 2025-11-08T17:19:23Z       NA        NA
5    Di  910           No Hastings 2025-11-09T16:15:25Z       NA        NA
6    Di  817          Yes Hastings 2025-11-09T17:49:47Z       NA        NA
           Notes
1 Did not tag NA
2          NA NA
3          NA NA
4          NA NA
5          NA NA
6          NA NA
```

Goal:

Graph each continuous quantitative variable versus head length. Get rid of outliers. Decide which variables are redundant and do not change much with species. These can be removed from the PCA later.

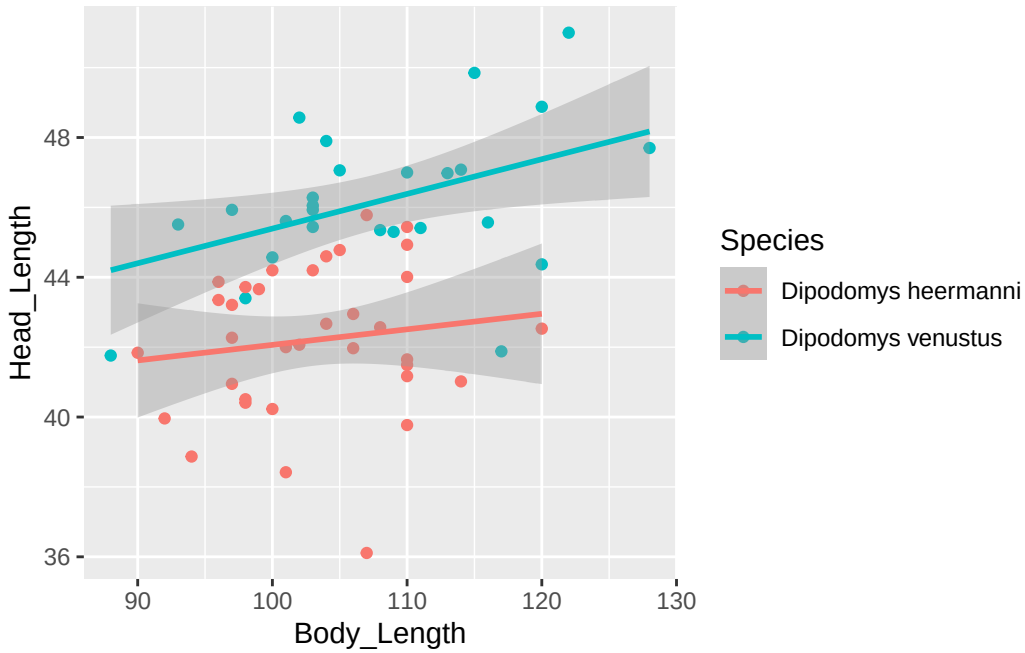
1. Assess head length for any outliers

```
#In this plot we can see a head_length value of about 36 which could be an outlier??
df %>%
  ggplot(aes(y= Head_Length, x= Body_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 37 rows containing missing values or values outside the scale range
(`geom_point()`).
```

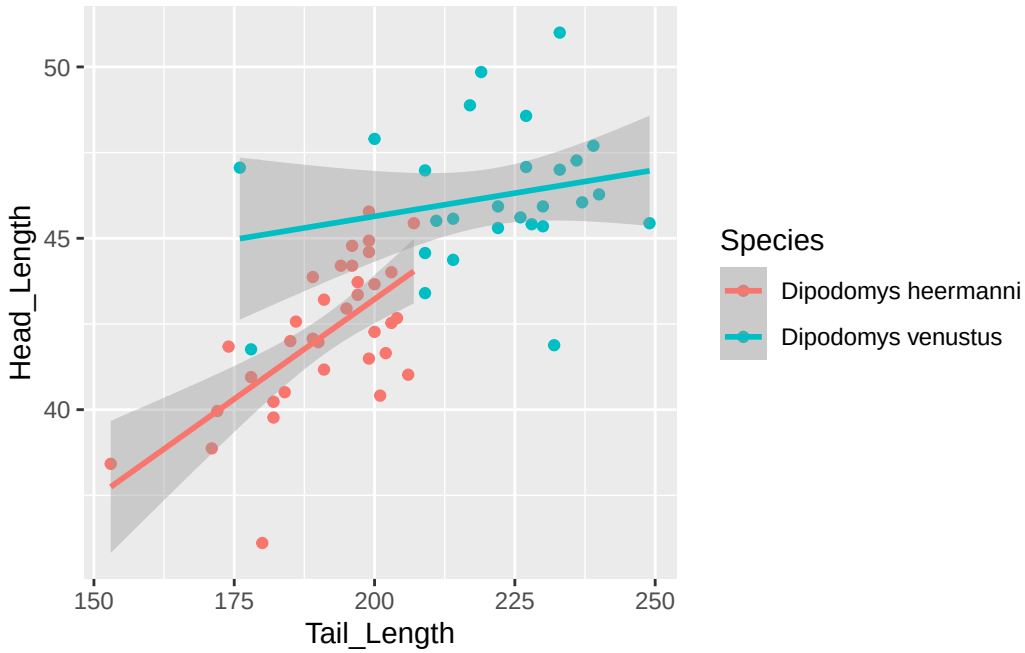



```
#Plotting versus tail length, we can see it is far out here as well
df %>%
  ggplot(aes(y= Head_Length, x= Tail_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 36 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 36 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Remove head length outlier

```
#Create new copy data frame
df1 <- df
#Remove possible outlier
df1$Head_Length[df$Head_Length < 37] <- NA

#Write new csv with the new change
#write_csv(df1, "10.8.2026KratCleanDataRMC.csv")

#Load in clean data set without head length outlier (hlo)
df.hlo <- read.csv("10.8.2026KratCleanDataRMC.csv") #w/o head length outlier 36 m
```

2. Head length outlier may not be an outlier

Our data demographics

All up to date, data points from Summer 2026

36 heermanni, 24 venustus

```r
#Remove rows with the most NAs in continuous, quantitative variables or where most measureme
df <- df[-c(6, 10, 11, 12,13, 14,15, 16, 19, 20, 21, 22, 23, 24, 25, 34, 36, 37, 38, 39, 51,
table(df$Species)
```

```
Dipodomys heermanni  Dipodomys venustus
                 36                  24
```

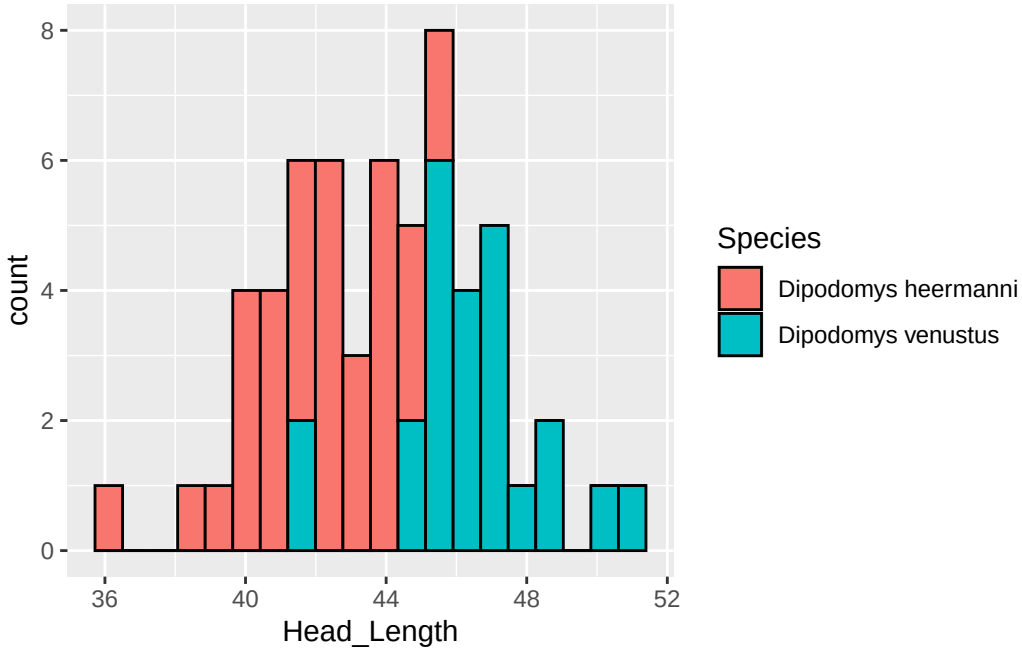
General size frequency distributions for Head Length

With all data points

Option here between a histogram and a density distribution. I like the density one better for looks, but probably tells you less if you don't know about the data for sample size, species counts and the actual amount the density assesses.

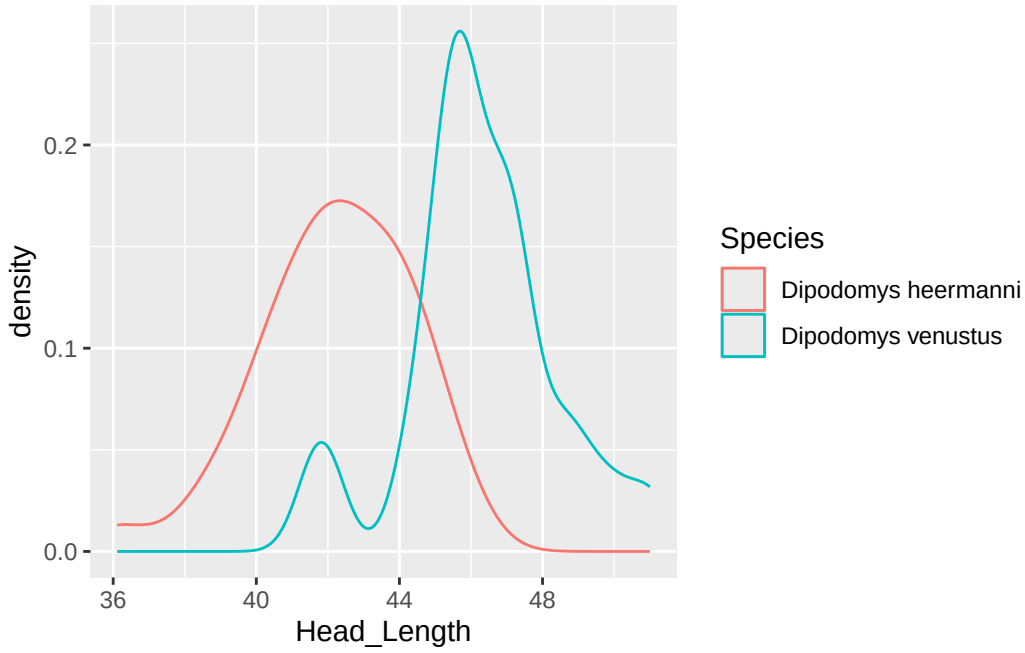
```r
#Create a histogram of the head length, separating by species
df %>%   ggplot(aes(x= Head_Length, fill =  Species))+ geom_histogram(bins = 20, color = "bla
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_bin()`).
```



```
#Create a density, smooth line frequency distribution, separating by species
df %>%   ggplot(aes(x= Head_Length, color =  Species))+ geom_density()
```

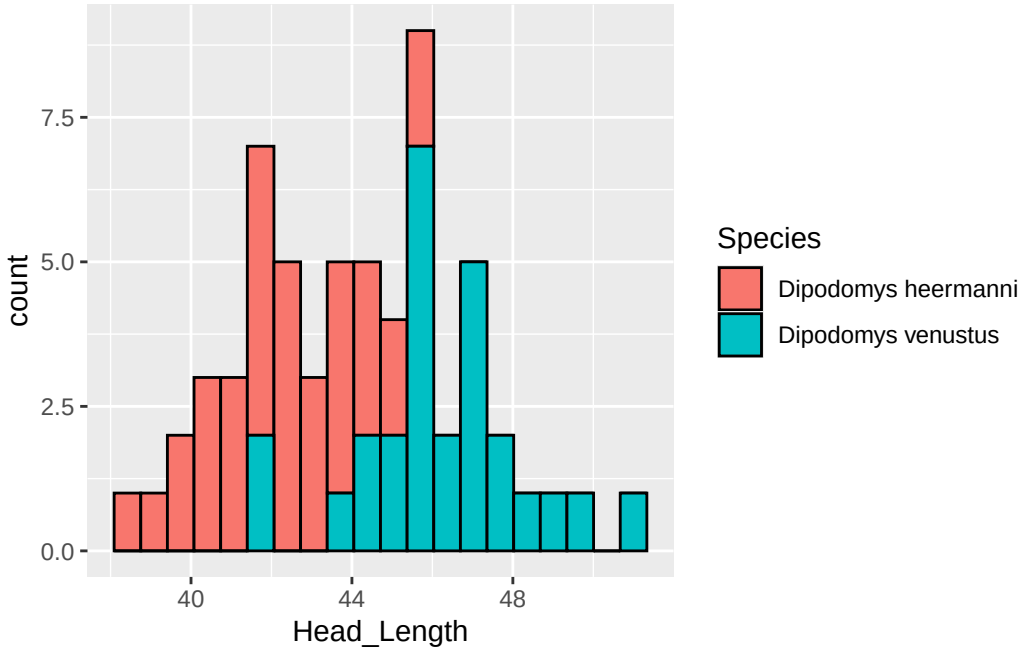
```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```



Without head length outlier (< 37 mm)

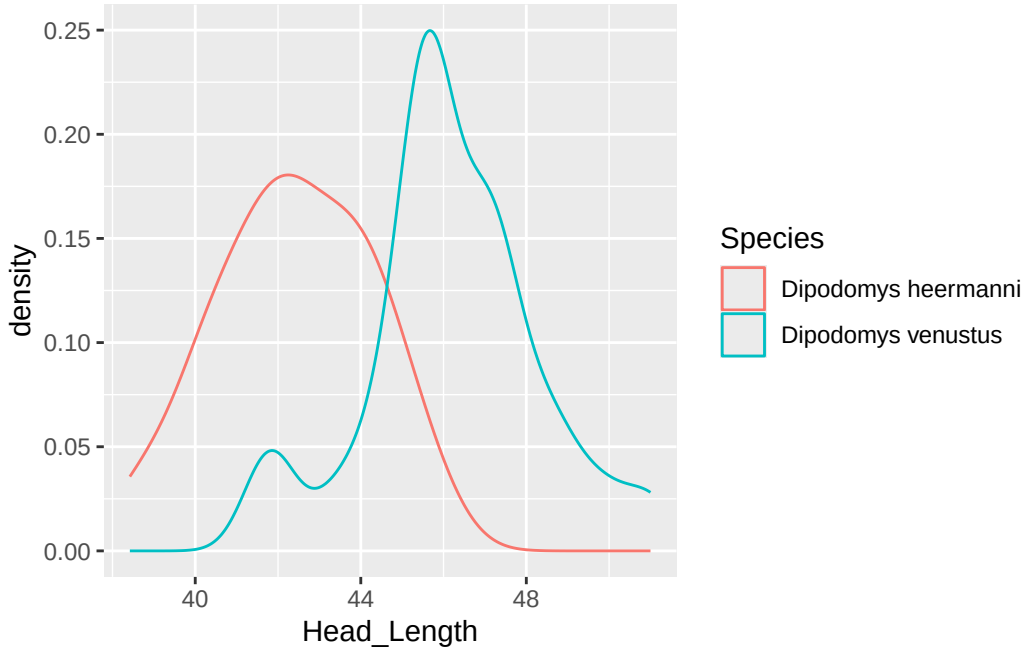
```
#Create a histogram of the head length, separating by species without the outlier
df.hlo %>%   ggplot(aes(x= Head_Length, fill =  Species))+ geom_histogram(bins = 20, color =
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_bin()`).
```




```
df.hlo %>%   ggplot(aes(x= Head_Length, color =  Species))+ geom_density()
```

```
Warning: Removed 37 rows containing non-finite outside the scale range
(`stat_density()`).
```



Tail is left out of the trait distribution when the head length value <37 mm is not included. Does this mean its actually an outlier?

Old spring data showing the outlier (what we looked at in meeting 10/8/26)

The more data collected over the summer filled in the gaps of the distribution that had been missing in this spring data analysis. This head length value <37 mm may not be an outlier after all??

Observation demographics of old data: 27 heermanni, 14 venustus

```
#Read in old data from the Spring, May 2026
df.old <- read.csv("final_krat_data_2026.csv")

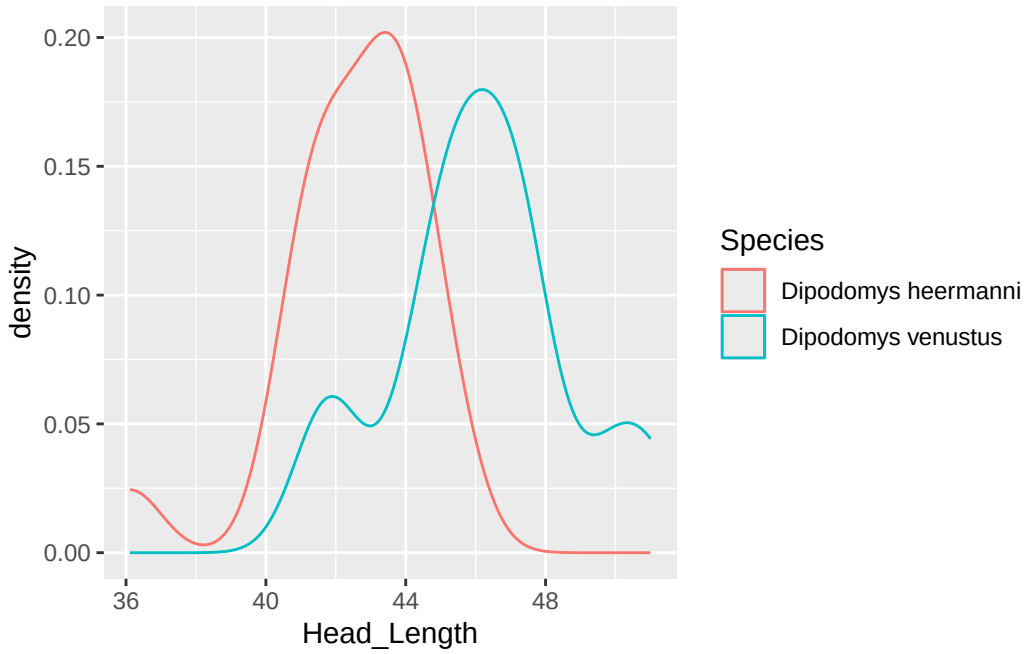
#Remove rows where most data was not taken (recaps mostly)
df.old <- df.old[-c(6, 10, 11, 12,13, 14,15, 16, 19, 20, 21, 22, 23, 24, 25, 34, 36, 37, 38,

#Have 27 heermanni, 14 venustus in this analysis from the spring before summer data was taker
table(df.old$Species)
```

```
Dipodomys heermanni  Dipodomys venustus
                 27                  14
```

```
#Create a density, smooth line frequency distribution, separating by species
df.old %>%
  ggplot(aes(x= Head_Length, color =  Species))+
geom_density()
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_density()`).
```



This might be explained by more juveniles being trapped in the early summer. If you look at the seasonal frequency distributions (in other analysis) this shows some pretty good evidence of size differences by seasons in Dipodomys heermanni.

3. Why head length as a size proxy?

Why head length and not body length as a size proxy?

Head length is being used due to its value as a general size indicator and also because it is a less coarse measurement than body length. When measuring small mammals a different scruff can change the body length and it is really a general estimate. The head length was taken with a caliper and is highly precise.

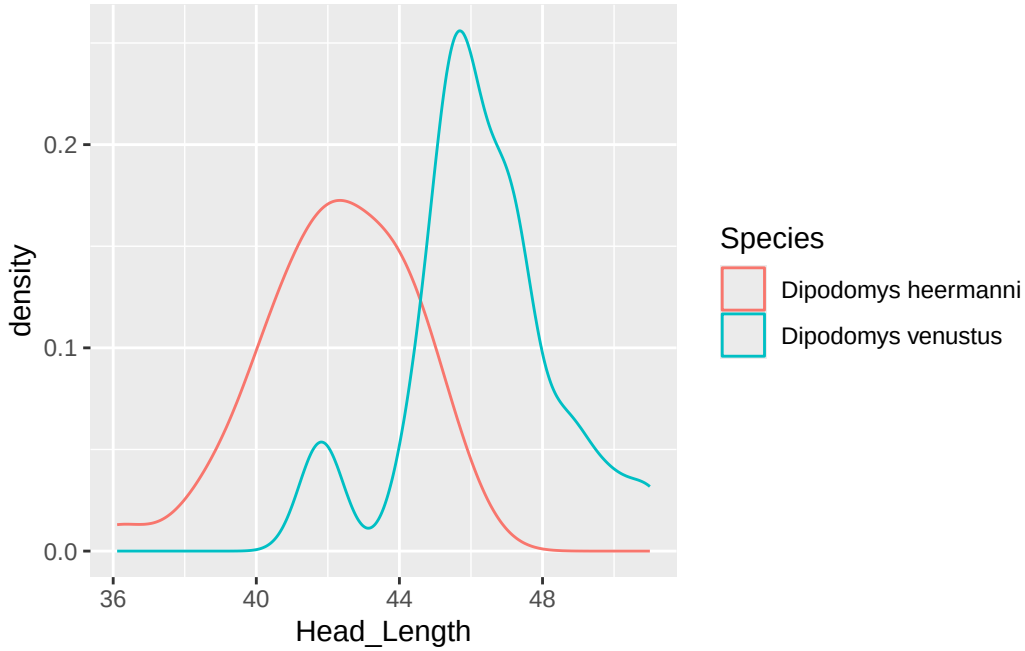
Additionally, we will see this later in data, body length is almost entirely overlapping between D. heermanni and D. venustus from our sample meaning body length would not help to differentiate species in a PCA analysis and head length would be a better size proxy to help find separation between the two.

Explore body length vs. head length as a measure of general size

Frequency distributions of Head length

```
#Create a density, smooth line frequency distribution, separating by species
df %>%
  ggplot(aes(x= Head_Length, color =  Species))+
geom_density()
```

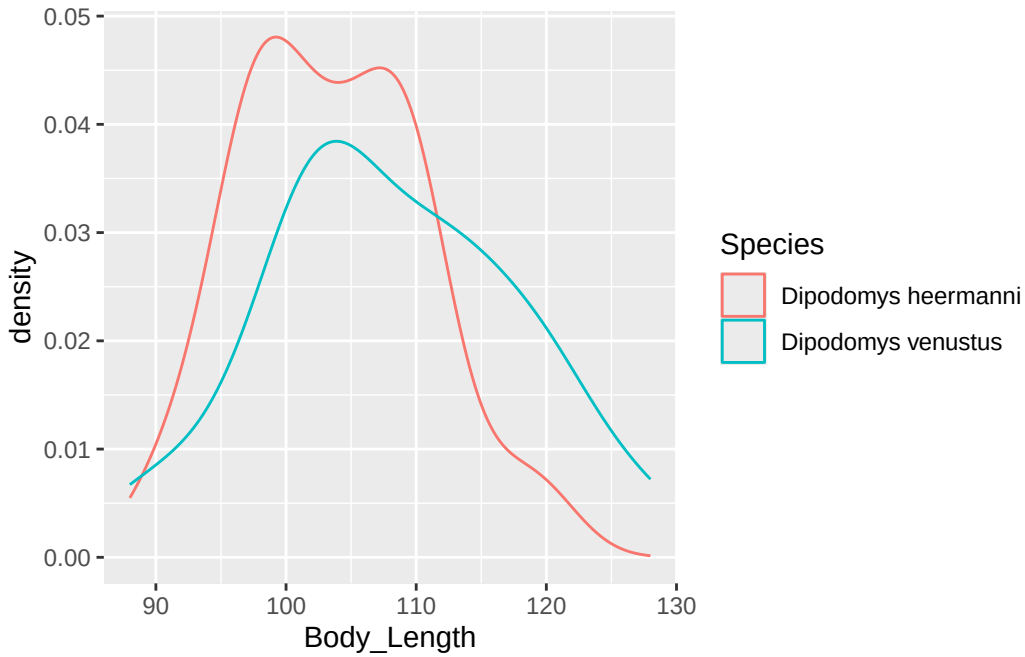
```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```



Frequency distributions of body length

```
#Create a density, smooth line frequency distribution of body length, separating by species
df %>%
  ggplot(aes(x= Body_Length, color =  Species))+
geom_density()
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_density()`).
```



Interpretation:

Body length seems to be very similar and nearly entirely overlapping on both species, but head length is different, but overlapping. This could mean that the ratio of head to body length is different on these animals.

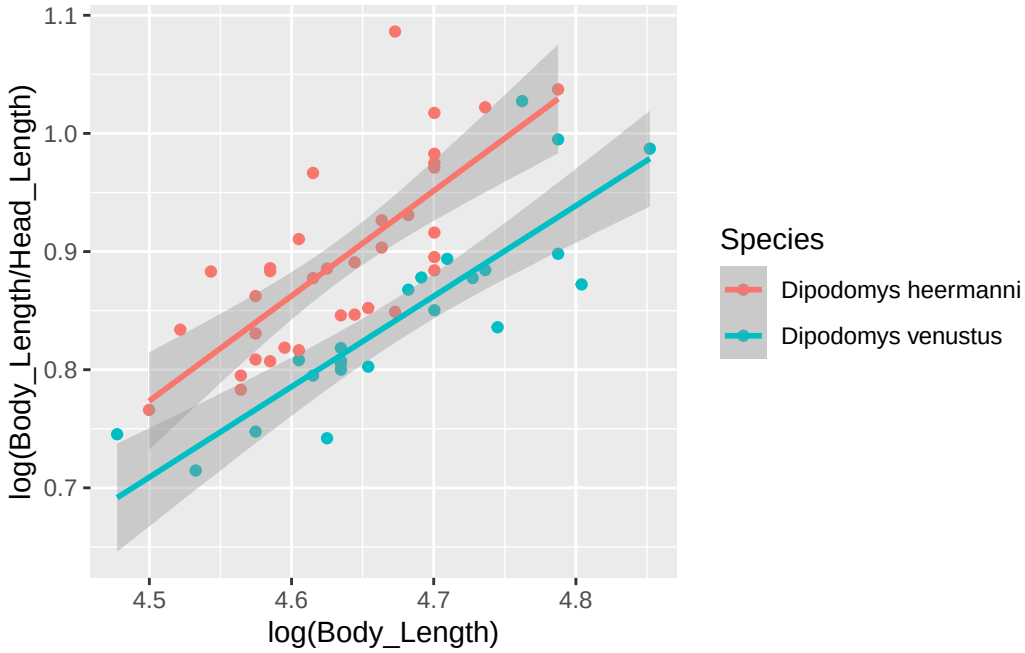
Ratio of body length to head length versus body length

```
#Show data of smooth linear regression plot
df |>
  ggplot(aes(y= log(Body_Length/Head_Length), x= log(Body_Length), color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

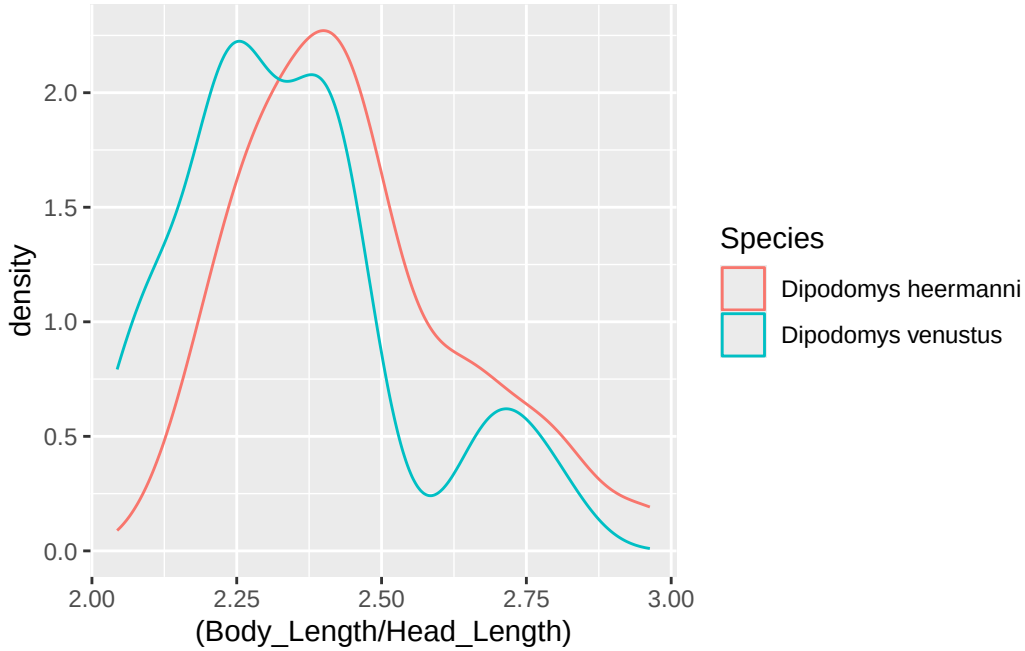
```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
#Show body length to head length ratio in a density frequency distribution plot
df %>%
  ggplot(aes(x= (Body_Length/Head_Length), color =  Species))+
geom_density()
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).
```



They don't seem to be too different and the body length to head length ratio would not help us to identify between these species very accurately. The head length on its own is more helpful than including body length. Consider excluding body length from PCA?

4. Graph each measurement versus head length

Only quantitative, continuous measurements will be graphed for outliers

```
#Here are all of the variables measured
colnames(df)
```

```
 [1] "OBJECTID"              "GlobalID"              "Collection_Date"
 [4] "Tissue_Sample"         "Hind_Limb_Length"      "Foot_Width"
 [7] "Wrist_Width"           "Forelimb_Length"       "Bolea_Head_Width"
[10] "Maxillary_Head_Width" "Head_Length"           "Finger_Length"
[13] "Hand_Width"           "Ear_Length"            "Toe_Length"
[16] "Foot_Length"          "Tail_Length"           "Body_Length"
[19] "Reproductive_Stage"   "Sex"                   "Life_Stage"
[22] "Weight_Animal_Only"   "Weight_Bag_Animal"     "Species"
[25] "Genus"                "Tag"                   "YN_Recapture"
[28] "Location"             "Date"                  "Latitude"
[31] "Longitude"            "Notes"
```

Quantitative, continuous measurements to try:

1) Body length

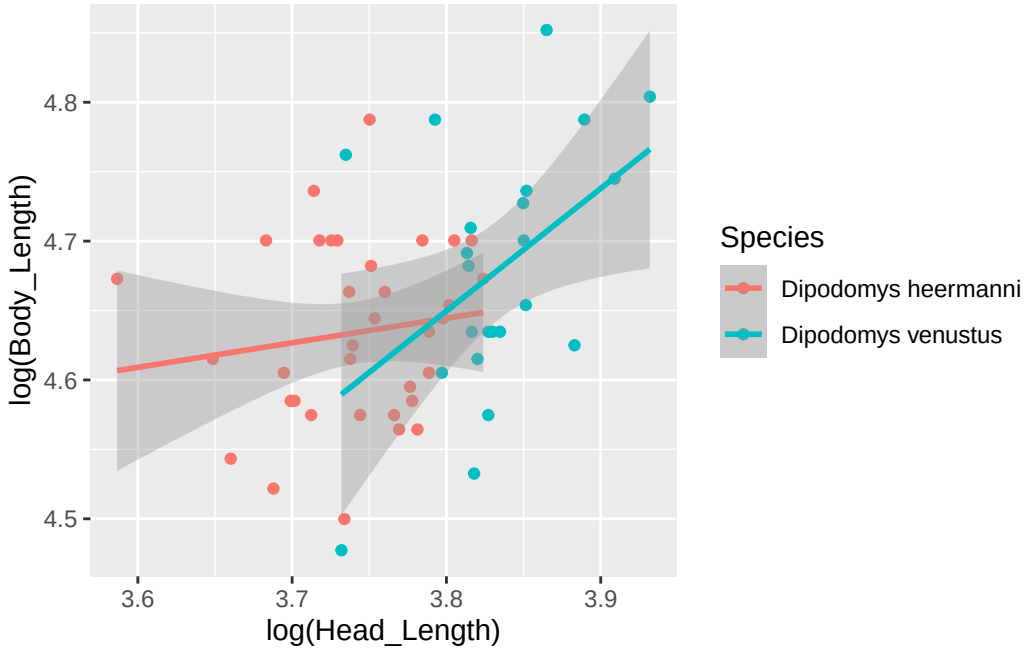
I'm trying logged versus normal and it doesn't seem to make a noticeable difference in the graphs. It only seems to get rid of real biological references like the millimeter units from the graph making it harder to understand.

```
#Body length vs head length as logged
df %>%
  ggplot(aes(y= log(Body_Length), x= log(Head_Length), color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

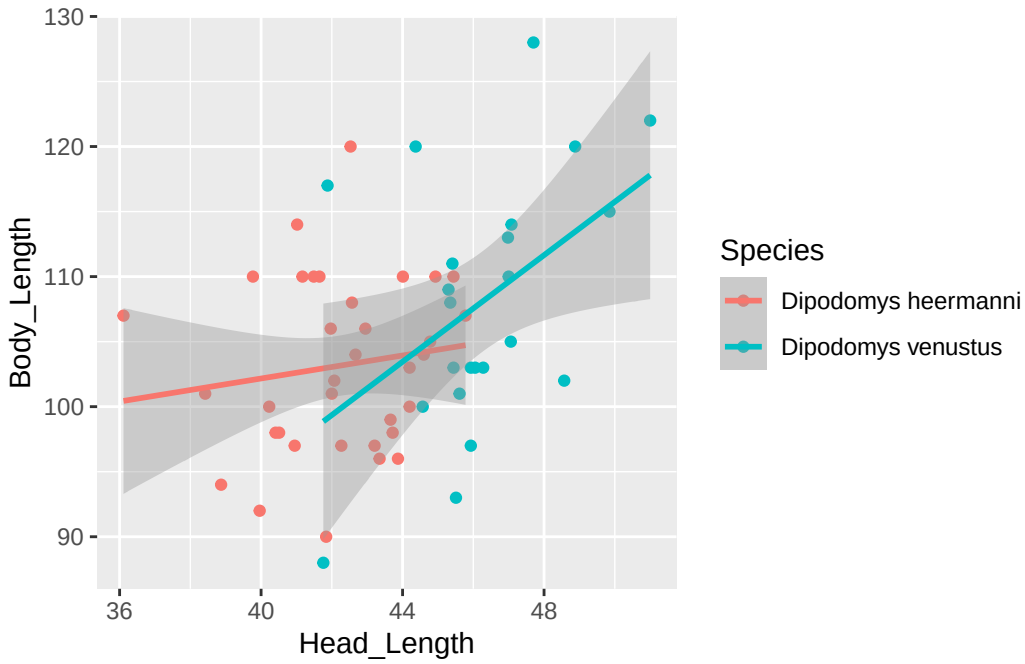
```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



```
#body length vs head length, not logged
df %>%
  ggplot(aes(y= Body_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range (`stat_smooth()`).
Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



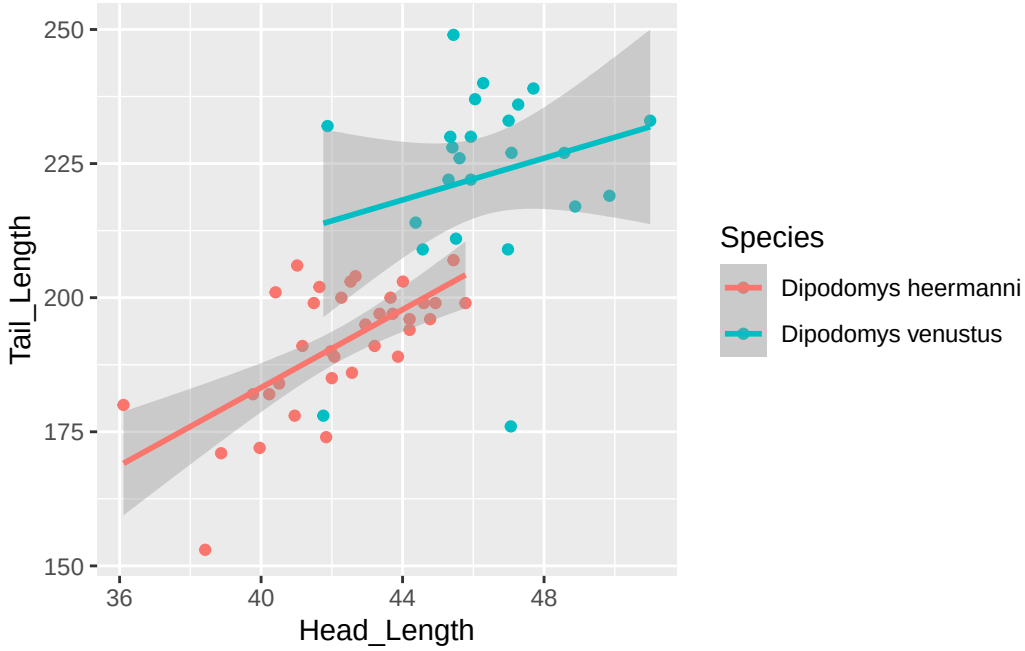
2) Tail length

```
#Tail length vs head length graph
df %>%
  ggplot(aes(y= Tail_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

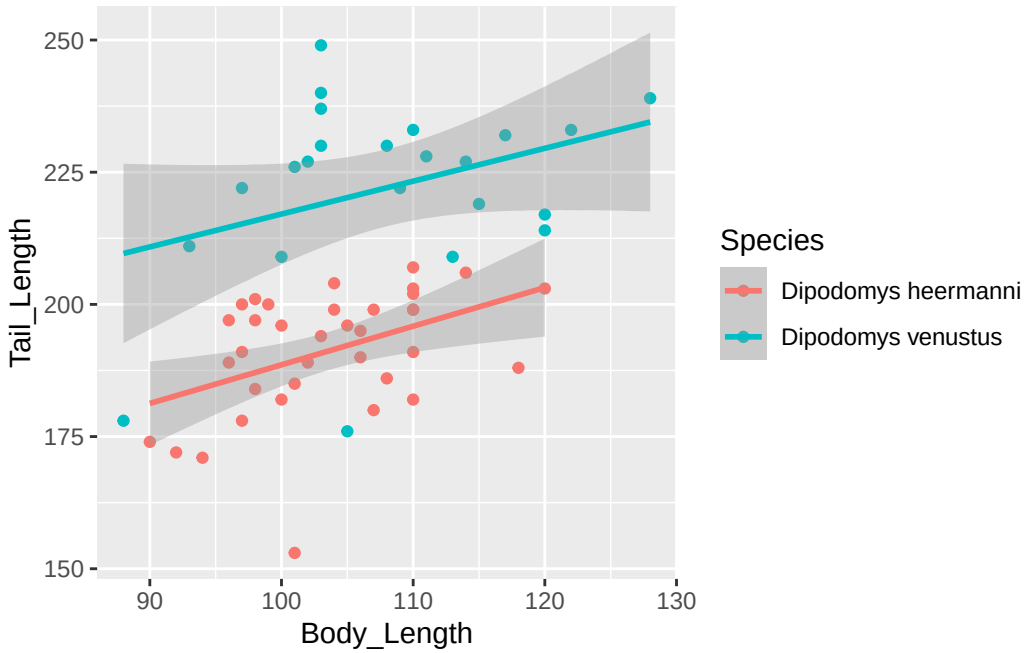
```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```




```
#Tail length vs head length
df %>%
  ggplot(aes(y= Tail_Length, x= Body_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range (`stat_smooth()`).
Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

There seem to be 2 outliers in the D. venustus with tail lengths $<$ 200 mm. When trapping its very possible that those kangaroo rats had the tip of its chopped or bitten off shortening it. I actually do remember some individuals like that at Hastings.

I looked back at these two individuals with tail lengths of ~178 mm, they were both trapped on the same day 6/22/2025 which early summer when we expect to see grown up juvenile-ish young adult kangaroo rats of the season. One of these individuals, tag 108 was captured again on 11/08/2025 and had a longer tail of 211 mm. This proves that these outliers really just represent younger individuals that are still growing.

This could be the same case for the D. heermanni with a tail length $<$ 160 mm. Looking back in the notes of the data set the individual, tag F002 trapped on 6/30/2026 is a juvenile, proving that these outliers are due to lifestage and represent important seasonal and lifestage variation.

Interpretation:

This data hints that young adult kangaroo rats of the season may not differ greatly from other adults in head length or body length, but do differ greatly in tail length. This means tail length could have the potential to be a good indicator of age and life-stage in kangaroo rats and picking out the young of the year in data sets.

3) Weight (animal only)

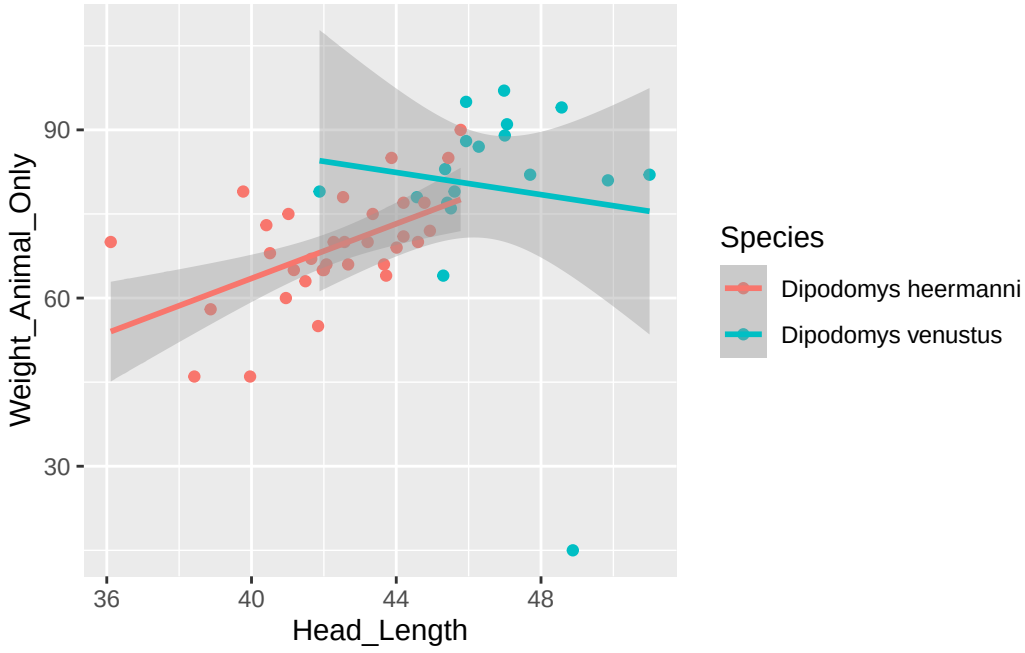
Side note: When I was working on getting rid of NAs for weight values, for some values we had a bag + animal weight but no bag weight, and I could estimate the bag weight for about 5 values to fill up this data. TBD

```
#Weight of animal versus head length graph
df %>%
  ggplot(aes(y= Weight_Animal_Only, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

There seems to be 1 really crazy outlier for the animal weight where it is < 30 grams which is insane. This was D. venustus individual, tag 843 that has a animal only weight of 15 grams. This is a data taker error, as there are other data taker errors in this observation. This value of 15 grams is pretty much how much the bags weigh and the bag+animal weight is shown to 115 grams, so I can be very sure this animal is actually 100 grams not 15g.

(This becomes clear once this was graphed again later, but there is another possible outlier for D. venustus that is less <70 g. Looking through notes, this D. venustus individual, tag B135, trapped 6/23/2026 has a weight of 64 g and is a young adult female. This difference can largely be explained by a different lifestage in our data. (interestingly the tail is really long 222 mm disproving the last point of short tail = young, atleast in this case)

Fix outliers:

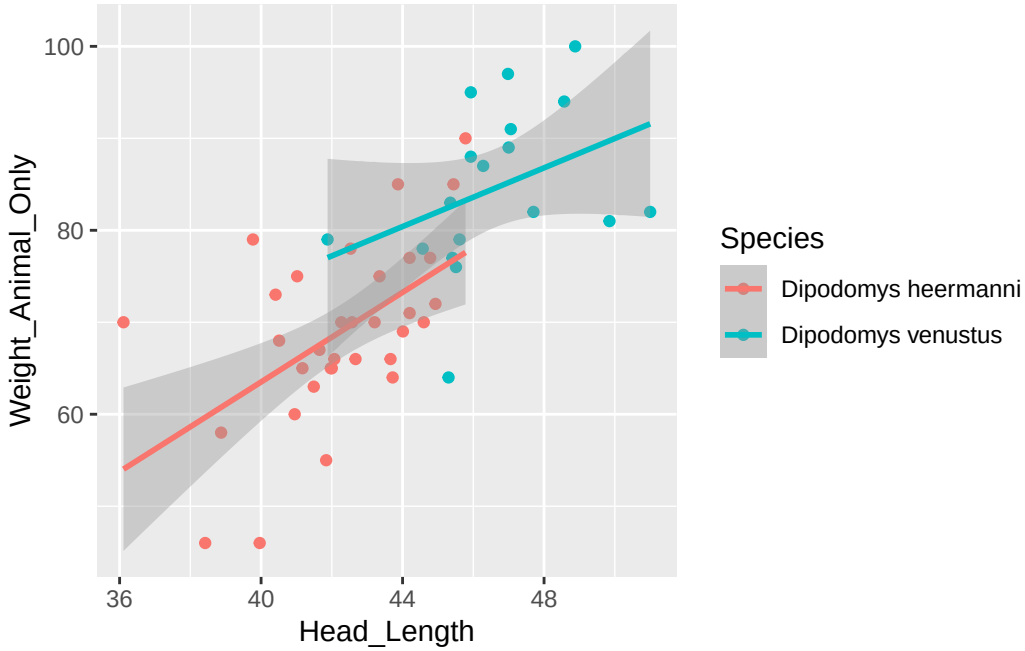
```
#Create new dataframe df.o (dataframe.outliers) which will include all my edits to remove ou
df.o <- df
#Fix outlier to true value, explained why above
df.o$Weight_Animal_Only[df$Weight_Animal_Only == 15] <- 100

#Plot without the outliers
df.o %>%
  ggplot(aes(y= Weight_Animal_Only, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

This data shows that animal weight and head length together are very good indicators of species between these two!! With some weights and head lengths that overlap and can be tricky. Around 75 - 85 g weight and 42 - 46 mm head length are the overlaps which species cannot be told here.

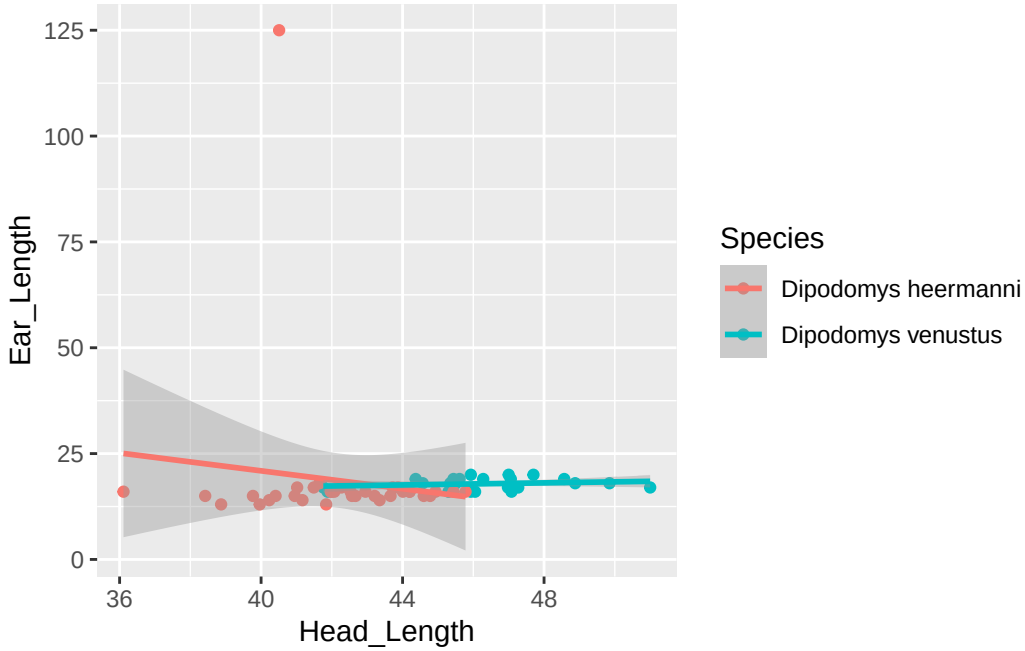
4) Ear length

```
#graph the ear length vs head length
df %>%
  ggplot(aes(y= Ear_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

Theres a crazy outlier for ear length that is 125 mm, which is likely explained as a typing error. That is likely a tail length or body length.

Fix outliers

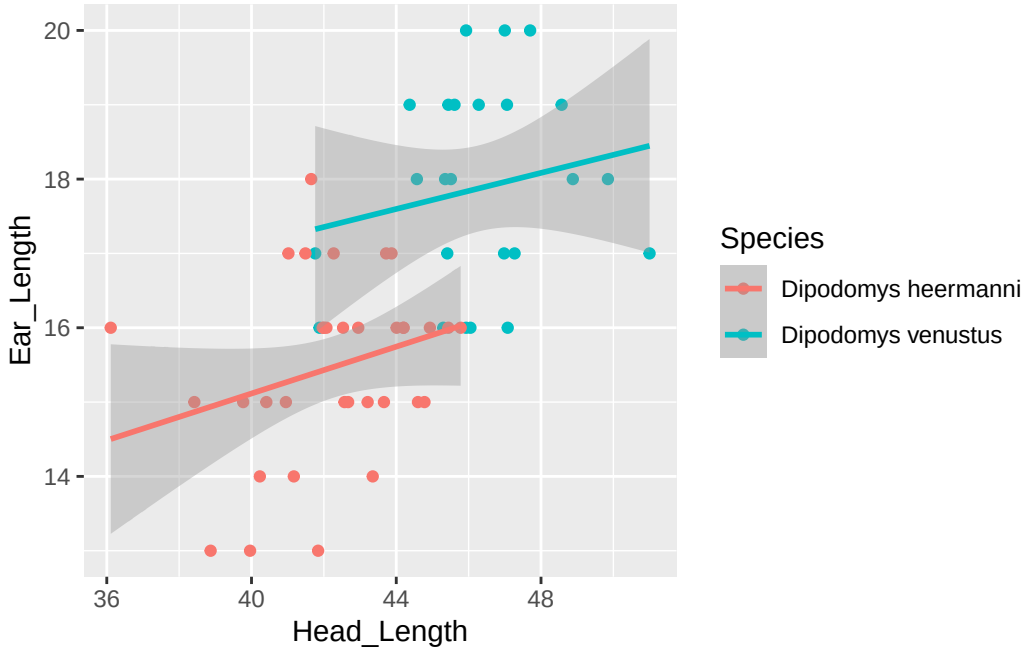
```
#Exclude any ear length values greater than 25 mm (already a crazy number for ear length here
df.o$Ear_Length[df$Ear_Length > 25] <- NA

#Regraph with outliers removed
df.o %>%
  ggplot(aes(y= Ear_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

The ear length and head length seem to be pretty good at distinguishing between species for most values. Between 16-18 mm of ear length and 42-46 mm head length we cannot be sure of species.

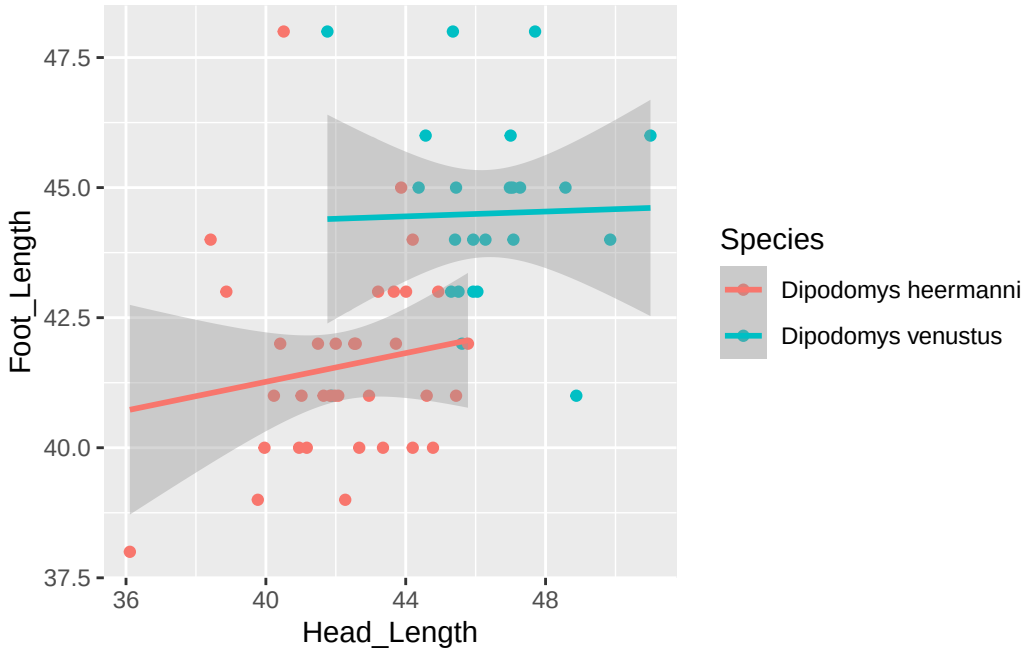
5) Foot length

```
#Graph foot length vs head length
df %>%
  ggplot(aes(y= Foot_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

I'm not really sure with this graph since more than half of the values are outside of the confidence intervals and the data is consistently super spread.

Looking at notes for D. heermanni individual with footlength of 48 mm. Individual was tag #12 caught in 7/12/2025, and this is the same individual which had a data entry error for the ear of 125 mm (crazy insane totally data taker fault), this could suggest the data taker potential may not have typed this correctly???

```
#Show individual heermanni with 48 mm foot length
df |> filter(Tag == 12)
```

```
  OBJECTID                                GlobalID      Collection_Date
1      100 {196AB800-A8A8-4B37-8678-00BB397A9503} 2025-07-12T18:14:49Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-10            45.45         NA       30.37           35.36
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1               NA                30.23       40.51             9          7
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1        125         14          48         184          98   Not Reproductive
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal             Species
1 Female      Adult                 68                NA Dipodomys heermanni
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di  12           no     FONR 2025-07-12T18:07:12Z 36.66282  -121.753
               Notes
1 Tissue sample 10 NA
```

```
#Get rid of foot length value of tag #12 which had 48 foot as heermanni
df.o$Foot_Length[df$Tag == 12] <- NA

#Check that it worked -- it did!!
df.o$Tag[df.o$Foot_Length == 48]
```

```
[1] "910" NA    "108" NA
```

Looking at the venustus who had a footlength of 48 mm on the graph.

Tag #910 has a foot length of 48 mm, its other measurements like 239 mm tail and 20 mm ear length suggest this was a large individual in general and may represent the largest range.

Tag #108 has been caught multiple times and measured twice. First time it was measured 6.22.2025 it was 48 foot and then 6 months later 11. 8. 2025 it was measured as 43 foot. I dont think this is biologically possible and the 48 foot is likely a data taker error… maybe it was actually 38 mm??? But for now I will get rid of this measurement.

```
#Show tag of the ones with 48 mm foot -- which are 910, 108 (12 has already been dealt with)
df$Tag[df$Foot_Length > 47.5]
```

```
[1] "910" NA    "108" "12"
```

```
#Show tag 910
df |> filter(Tag == 910)
```

```
  OBJECTID                                GlobalID      Collection_Date
1        5 {B464F9EF-9309-4CA1-8F9C-1D45FF69CA99} 2025-11-09T16:19:56Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1          <NA>             52.9       5.57        2.74           36.15
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1            20.51                22.52        47.7            10          8
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         20         16          48         239         128   Non-Reproductive
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal            Species
1 Female      Adult                 82                105 Dipodomys venustus
  Genus Tag YN_Recapture Location                 Date Latitude Longitude Notes
1    Di 910           No Hastings 2025-11-09T16:15:25Z       NA        NA NA NA
```

```
#Show tag 108
df |> filter(Tag == 108)
```

```
  OBJECTID                                GlobalID      Collection_Date
1        2 {A720CCF9-2B13-4707-B8D2-426CE01002E3} 2025-11-08T16:19:58Z
2       74 {C20A9398-35A6-48E9-AE2A-2DFFF20CA52D} 2025-06-22T16:50:27Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-22            48.38       5.34        3.03           34.76
2          <NA>               NA         NA        3.58           34.60
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1             23.8                21.52       45.51             8          7
2               NA                30.60       41.76            NA          7
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         18         15          43         211          93            Unknown
2         17         NA          48         178          88               <NA>
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal            Species
1 Unknown      Adult                 76                94 Dipodomys venustus
2  Female      Adult                 NA                NA Dipodomys venustus
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di 108          Yes Hastings 2025-11-08T16:20:10Z       NA        NA
2    Di 108         <NA> Hastings 2025-06-22T16:39:34Z 36.38459 -121.5493
    Notes
1   NA NA
2 34.6 NA
```

```
#Tag 108 has multiple observations, lets get rid of just the 48 mm one
df.o <- df.o %>%
  mutate(Foot_Length = ifelse(Tag == 108 & Foot_Length == 48, NA, Foot_Length))
```

```r
#Check that it worked-- it worked!!
df.o |> filter(Tag == 108)
```

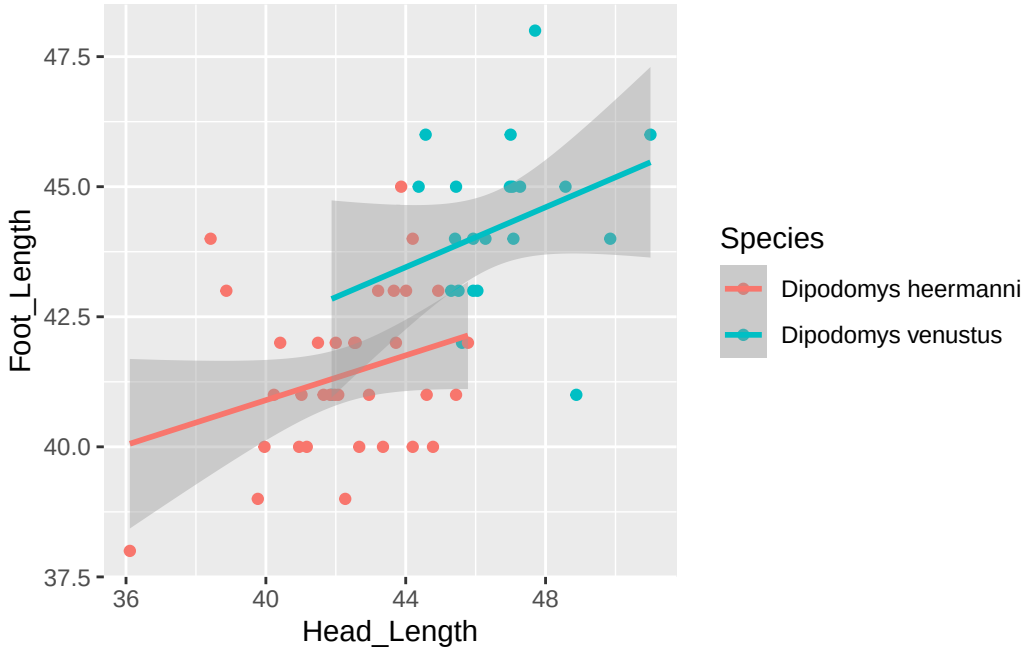
```
  OBJECTID                                GlobalID      Collection_Date
1        2 {A720CCF9-2B13-4707-B8D2-426CE01002E3} 2025-11-08T16:19:58Z
2       74 {C20A9398-35A6-48E9-AE2A-2DFFF20CA52D} 2025-06-22T16:50:27Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-22            48.38       5.34        3.03           34.76
2          <NA>               NA         NA        3.58           34.60
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1             23.8                21.52       45.51             8          7
2               NA                30.60       41.76            NA          7
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         18         15          43         211          93            Unknown
2         17         NA          NA         178          88               <NA>
      Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal            Species
1 Unknown      Adult                 76                94 Dipodomys venustus
2  Female      Adult                 NA                NA Dipodomys venustus
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di 108          Yes Hastings 2025-11-08T16:20:10Z       NA        NA
2    Di 108         <NA> Hastings 2025-06-22T16:39:34Z 36.38459 -121.5493
    Notes
1   NA NA
2 34.6 NA
```

```r
#Regraph with outliers removed
df.o %>%
  ggplot(aes(y= Foot_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 4 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 4 rows containing missing values or values outside the scale range
(`geom_point()`).
```




```
#Now just tag # 910 is left since we decided to keep it
df.o |> filter(Foot_Length > 47.5)
```

```
  OBJECTID                                GlobalID      Collection_Date
1        5 {B464F9EF-9309-4CA1-8F9C-1D45FF69CA99} 2025-11-09T16:19:56Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1          <NA>             52.9       5.57        2.74           36.15
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1            20.51                22.52        47.7            10          8
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         20         16          48         239         128   Non-Reproductive
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal            Species
1 Female      Adult                 82               105 Dipodomys venustus
  Genus Tag YN_Recapture Location                Date Latitude Longitude Notes
1    Di 910           No Hastings 2025-11-09T16:15:25Z       NA        NA NA NA
```

Interpretation:

There are clear differences in mean foot length between these species, but this trait seems to be incredibly variable with a lot of overlap among them which would make it hard to distinguish based on foot length when looking at a single individual.

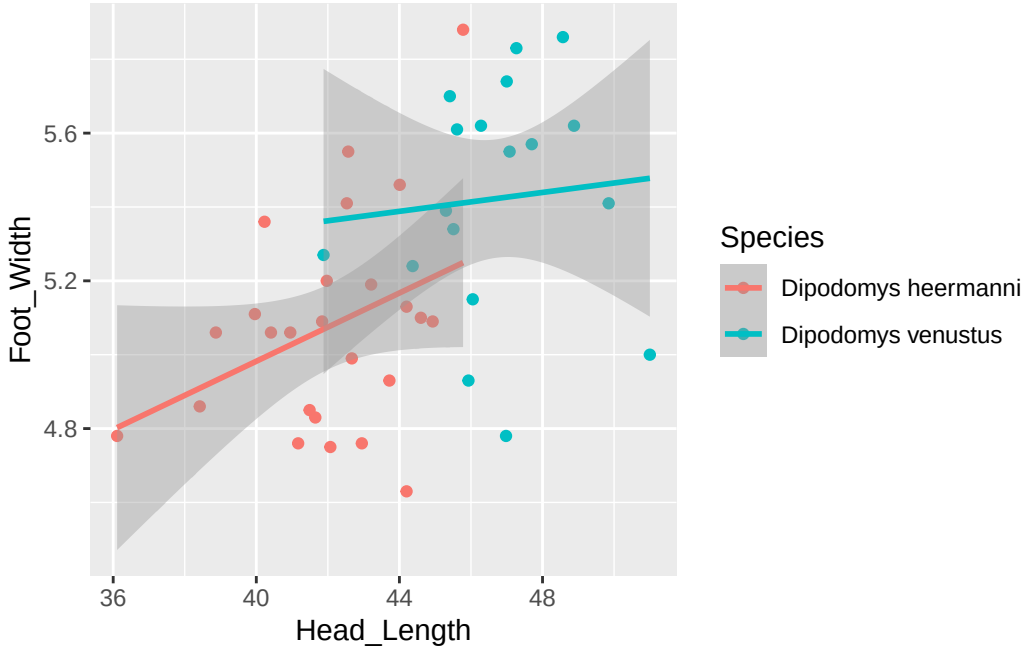
6) Foot width

```
#Graph foot width vs head length
df %>%
  ggplot(aes(y= Foot_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 17 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 17 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers + Interpretation:

D. venustus individual with ~6.5 mm foot width is pretty far out. Tag 843 has a foot width value of 6.25 mm when it was caught 4/11/2026 and then it has a lower foot width value of

5.62 mm when it was caught about 2.5 months later on 6/23/2026. It seems like this value should not decrease as the animal ages and could either suggest that this measurement is no good, or that maybe the muscle mass increases/ decreases over months??? (unlikely?)

Otherwise, foot width seems to show a clear difference in means between the two species which may be somewhat equivalent of a difference as foot length is. There is still a lot of overlap between the two.

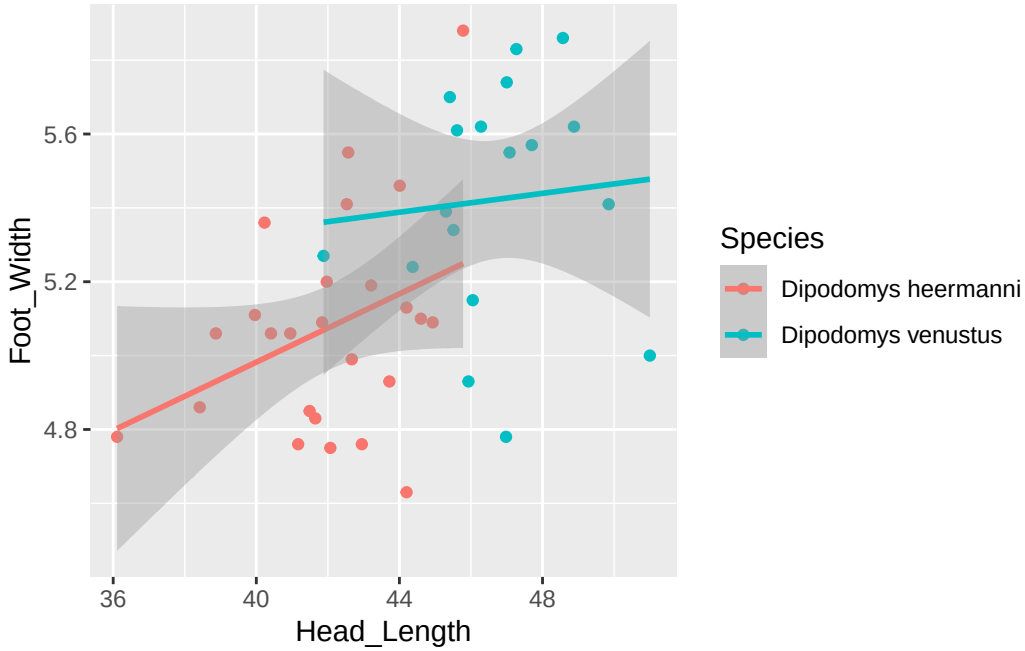
```
#Get rid of 6.25 mm foot width outlier
df.o$Foot_Width[df$Foot_Width > 6] <- NA

#Graph foot width versus head length without the outliers
df.o %>%
  ggplot(aes(y= Foot_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 17 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 17 rows containing missing values or values outside the scale range
(`geom_point()`).
```



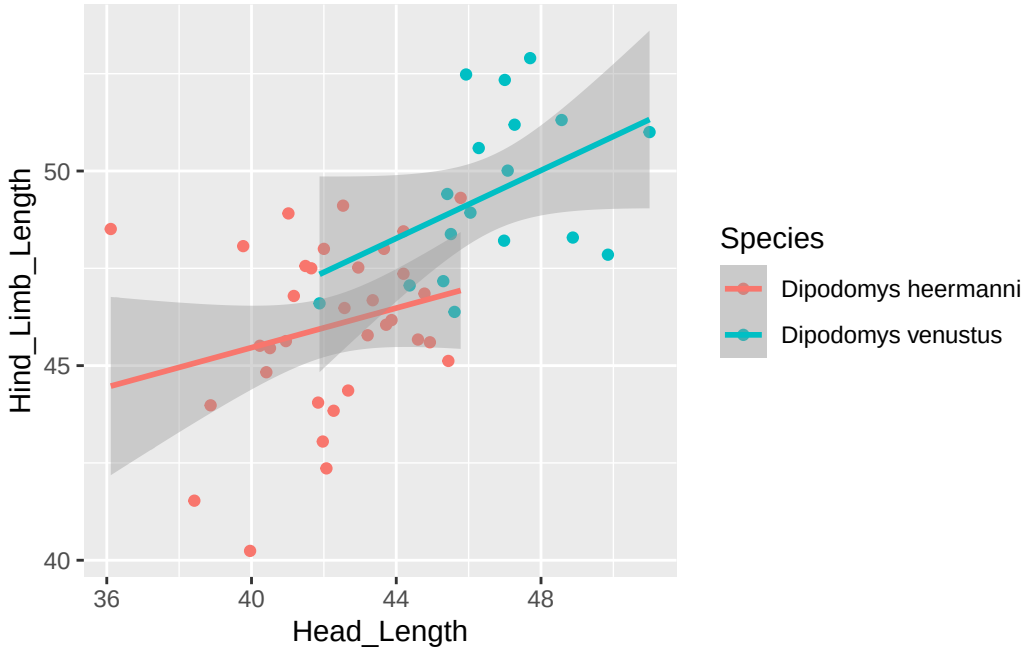
7) Hind limb length

```
#Graph hind limb length vs head length
df %>%
  ggplot(aes(y= Hind_Limb_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 8 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 8 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

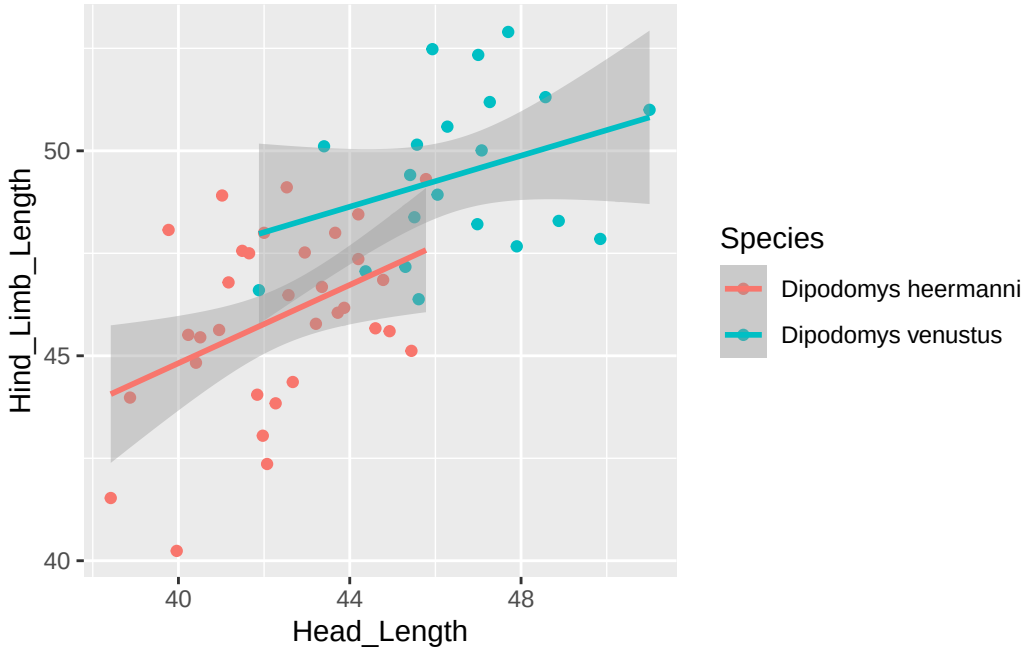
The data seems pretty spread, but the 36 mm head length heermanni seems to be a weird outlier. I had looked at this before, but I will try this graph without it to see if the linear regression slope changes. It does seem to change the slope in this case to be more of an equal slope with the venustus.

```
#Use the head length outlier subtracted dataframe for this
df.hlo |>
  ggplot(aes(y= Hind_Limb_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 44 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 44 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

This is a clear difference in slope that still overlap but are shown to have different y-intercept values showing a general difference between the species.

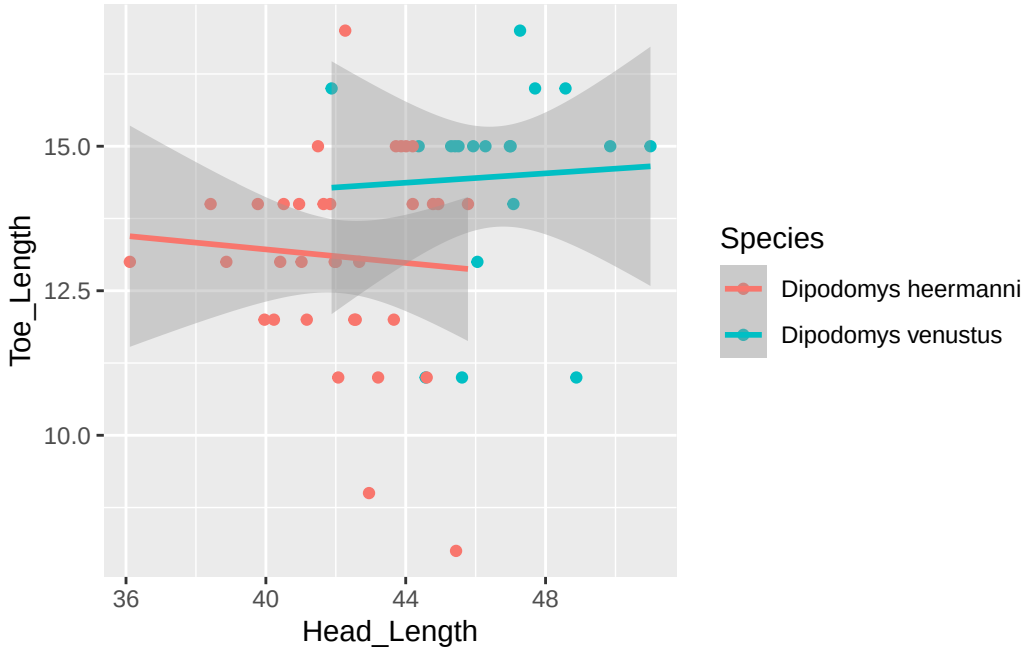
8) Toe length

```
#Toe length vs head length graph
df |>
ggplot(aes(y= Toe_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 7 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

Toe lengths less than 10 mm do seem to be dragging the slope down on toe length. Those should be taken out and were likely measured wrong or could represent juveniles with short feet already. They seem to have adult measurements that are pretty normal and average for kangaroo rats, so I will get rid of these outliers.

```
#Show outliers all measurements
df |> filter(Toe_Length < 10)
```

```
  OBJECTID                                GlobalID      Collection_Date
1      121 {F1100A79-8394-483B-8095-250519A2B38E} 2025-07-13T16:15:48Z
2      136 {0DC47928-75E9-412F-9E46-B45E2161492D} 2025-09-28T15:18:34Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-03            45.12         NA        3.67           28.33
2     RC2025-21            47.52       4.76        2.60           30.84
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1               NA                34.51       45.44             8          6
2            19.62                27.54       42.95             6          6
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         16          8          41         207         110          Lactating
2         16          9          41         195         106               <NA>
```

```
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal             Species
1 Female      Adult                 85                NA Dipodomys heermanni
2   <NA>       <NA>                 NA                NA Dipodomys heermanni
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di 114           no     FONR 2025-07-13T16:05:41Z   36.663 -121.7531
2    Di 123         <NA>     FONR 2025-09-28T15:09:23Z       NA        NA
                                                    Notes
1                                             Tissue 3 NA
2 2nd head width 19.62 tissue # RC2025-21 Forelimb length 30.84
```

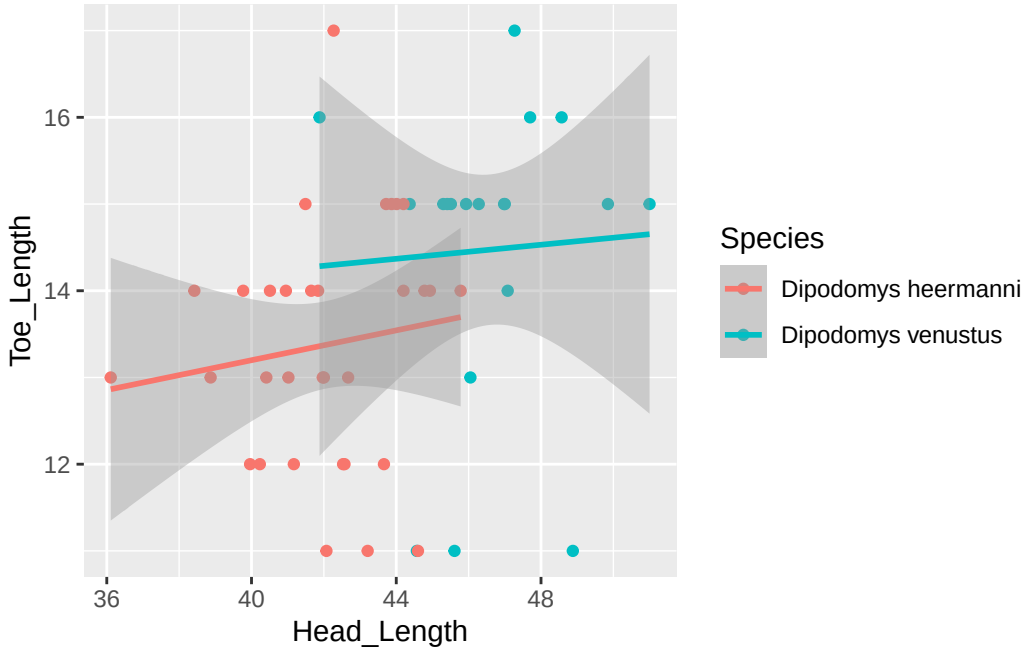
```
#Remove toe length measurements less than 10 mm
df.o$Toe_Length[df$Toe_Length < 10] <- NA

#Graph toe length vs head length without these outliers
df.o |>
  ggplot(aes(y= Toe_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

The toe lengths seem to be very spread and variable, but do have clear differences in mean.

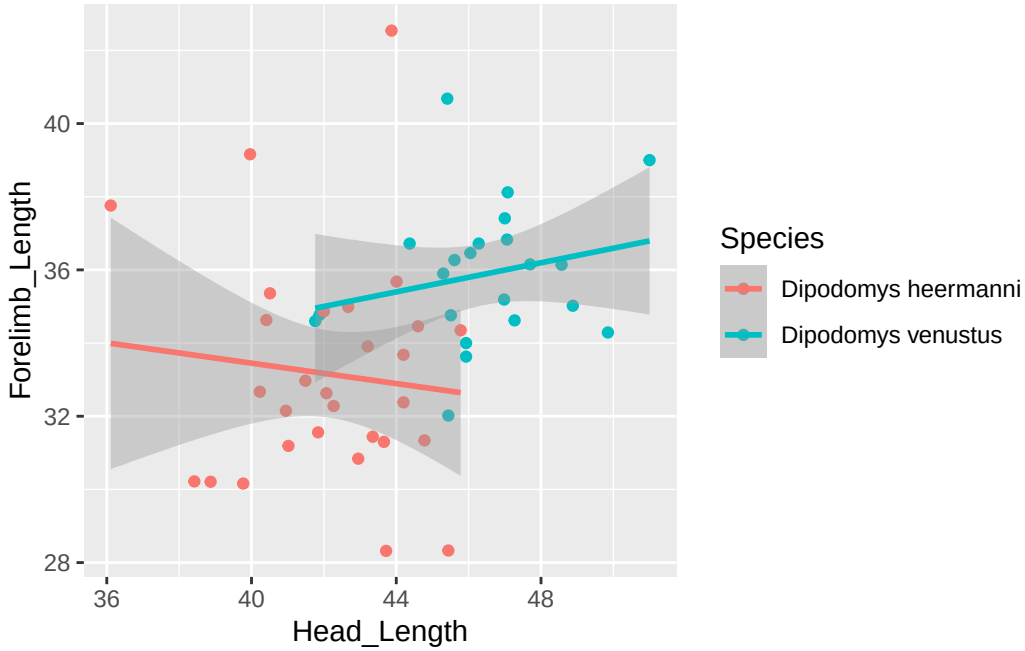
9) Forelimb length

```
#Graph forelimb length vs head length
df |>
ggplot(aes(y= Forelimb_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

The greater than 39.01 mm forelimb lengths seem like outliers. Looking at the observation of heermanni thats over 42 mm the finger length which is included in this measurement doesnt seem exceptionally long (8 mm) and I will remove it, along with other really big values of this.

The two lower values of venustus less than 29 mm also may be outliers??? The observations are normal with normal adult measurements, could just be low or could be a measurement error — once higher outliers are removed I will look back at them. With higher outliers removed they still look far out.

```
#Show observations with forelimb length greater than 42 mm
df |> filter(Forelimb_Length > 42)
```

```
  OBJECTID                                GlobalID      Collection_Date
1      119 {2FD6F21D-BAD6-4DAB-8C82-679B58CE154E} 2025-07-13T15:55:16Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-04            46.17         NA        2.68           42.54
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1               NA                28.65       43.87             8          8
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
```

```
1         17         15          45         189          96       Reproductive
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal             Species
1 Female      Adult                 85                NA Dipodomys heermanni
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di 112           no     FONR 2025-07-13T15:41:44Z 36.66331 -121.7535
       Notes
1 Tissue 4 NA
```

```
#Remove forelimb lengths greater than 42 mm
df.o$Forelimb_Length[df$Forelimb_Length > 42] <- NA

#Show observations with forelimb length less than 29 mm
df |> filter(Forelimb_Length < 29)
```

```
  OBJECTID                                GlobalID      Collection_Date
1      121 {F1100A79-8394-483B-8095-250519A2B38E} 2025-07-13T16:15:48Z
2      162 {8811F861-4BFE-492A-9032-F86B7FA72746} 2025-10-18T16:13:51Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-03            45.12         NA        3.67           28.33
2     RC2025-11            46.05       4.93        2.56           28.32
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1               NA                34.51       45.44             8          6
2            23.68                25.93       43.72             8          7
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1         16          8          41         207         110          Lactating
2         17         15          42         197          98               <NA>
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal             Species
1 Female      Adult                 85                NA Dipodomys heermanni
2   <NA>       <NA>                 64                NA Dipodomys heermanni
  Genus Tag YN_Recapture Location                 Date Latitude Longitude
1    Di 114           no     FONR 2025-07-13T16:05:41Z  36.6630 -121.7531
2    Di 119           no     FONR 2025-10-18T16:09:06Z  36.6626 -121.7528
                                                
```

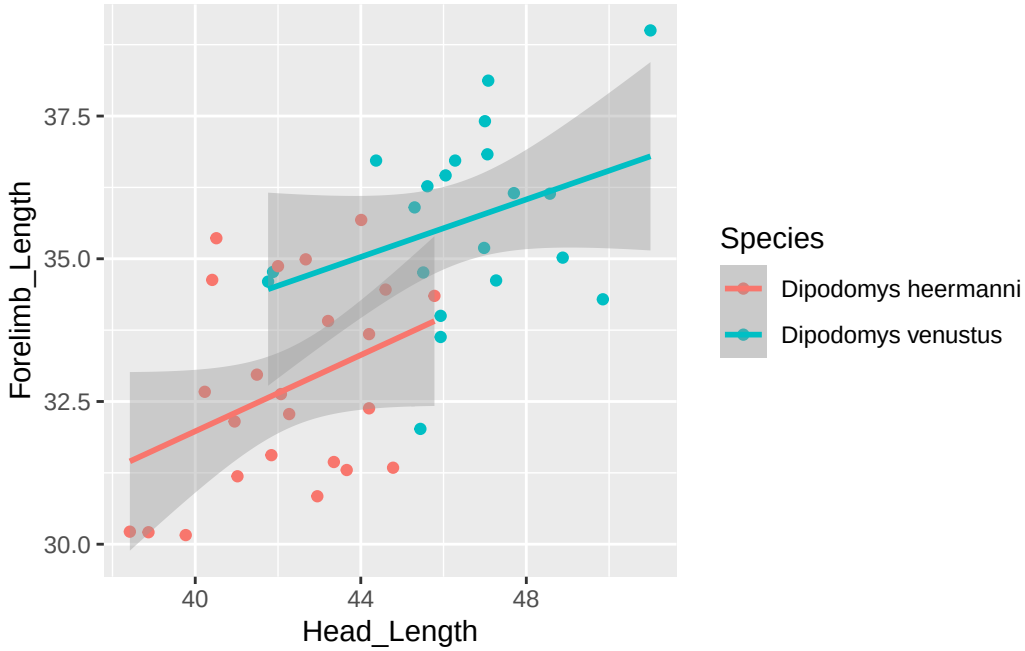
```r
#Remove forelimb lengths greater than 39.01 mm
df.o$Forelimb_Length[df$Forelimb_Length > 39.01] <- NA


#Graph forelimb length vs head length without outliers
df.o |>
  ggplot(aes(y= Forelimb_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 15 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 15 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

This graphs show pretty big differences in slope and mean for forelimb length and not very much overlap in confidence intervals, however there are a lot of overlapping values that exist for both species.

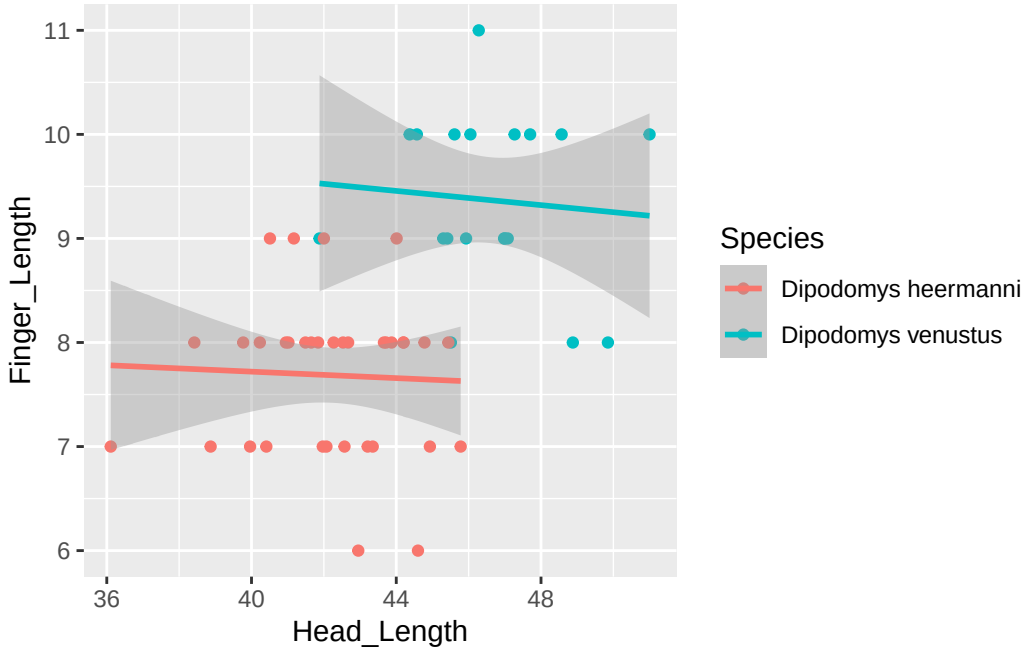
10) Finger length

```
#Graph the finger length versus head length
df |>
ggplot(aes(y= Finger_Length, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 6 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 6 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

There don't seem to be any crazy outliers except maybe the 6 mm and 11 mm values, but these may just represent the highest and lowest ranges.

Interpretation:

This graph shows some pretty flat slopes meaning finger length likely does not increase with head length or body size. This graph also shows only 2 overlapping values of finger length for these species. Additionally, the confidence intervals do not overlap and show some separation.

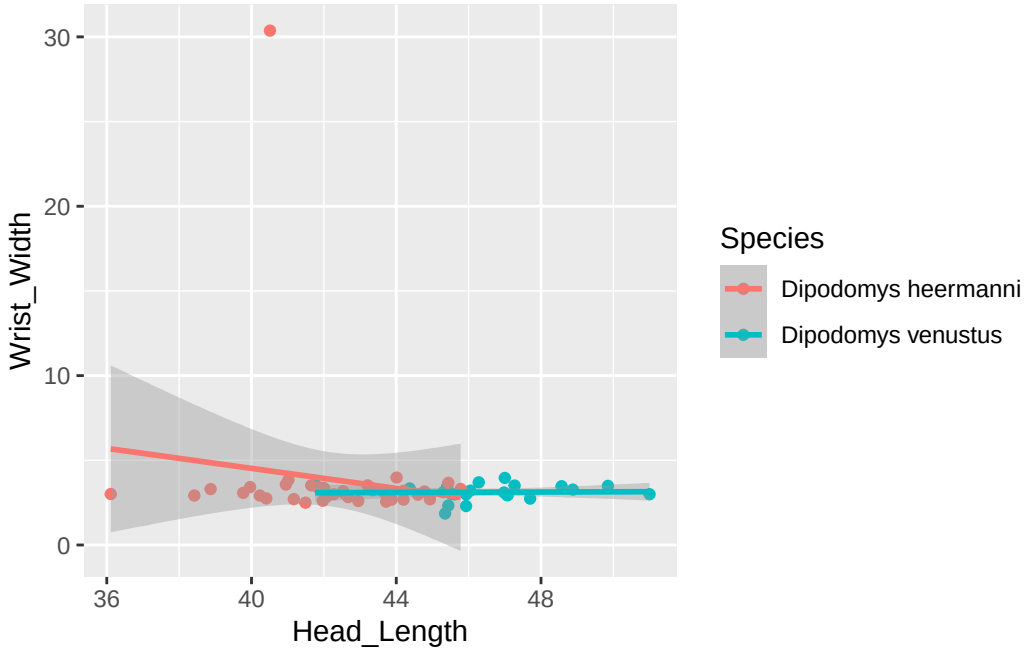
11) Wrist width

```
#Graph wrist width versus head length
df |>
ggplot(aes(y= Wrist_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outlier:

There is a crazy outlier here in the 30s for a measurement ranging between 2 -4 mm. This is definitely a data taker error. Is it possible the decimal point was just not added? – Tag # 12 The wrist width reads: 30.37 could this really be 3.37?? This is the same observation which had an ear length of 125 mm and a foot length of 48. This data taker was making a lot of mistakes and typos.

```
#Show observation with the outlier
df |> filter(Wrist_Width > 30)
```

```
  OBJECTID                                GlobalID      Collection_Date
1      100 {196AB800-A8A8-4B37-8678-00BB397A9503} 2025-07-12T18:14:49Z
  Tissue_Sample Hind_Limb_Length Foot_Width Wrist_Width Forelimb_Length
1     RC2025-10            45.45         NA       30.37           35.36
  Bolea_Head_Width Maxillary_Head_Width Head_Length Finger_Length Hand_Width
1               NA                30.23       40.51             9          7
  Ear_Length Toe_Length Foot_Length Tail_Length Body_Length Reproductive_Stage
1        125         14          48         184          98   Not Reproductive
     Sex Life_Stage Weight_Animal_Only Weight_Bag_Animal             Species
1 Female      Adult                 68                NA Dipodomys heermanni
  Genus Tag YN_Recapture Location                Date Latitude Longitude
```

```
1    Di  12           no     FONR 2025-07-12T18:07:12Z 36.66282  -121.753
                Notes
1 Tissue sample 10 NA
```

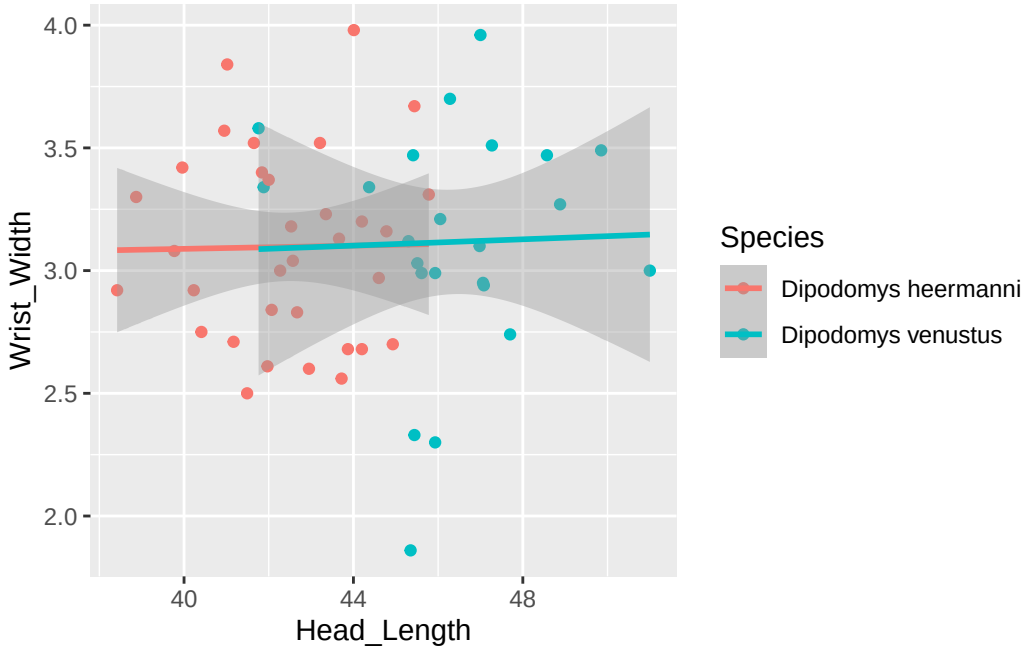
```r
#Remove outlier
df.o$Wrist_Width[df$Wrist_Width > 30] <- NA

#Graph without the outlier
df.o |>
  ggplot(aes(y= Wrist_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 4 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 4 rows containing missing values or values outside the scale range
(`geom_point()`).
```



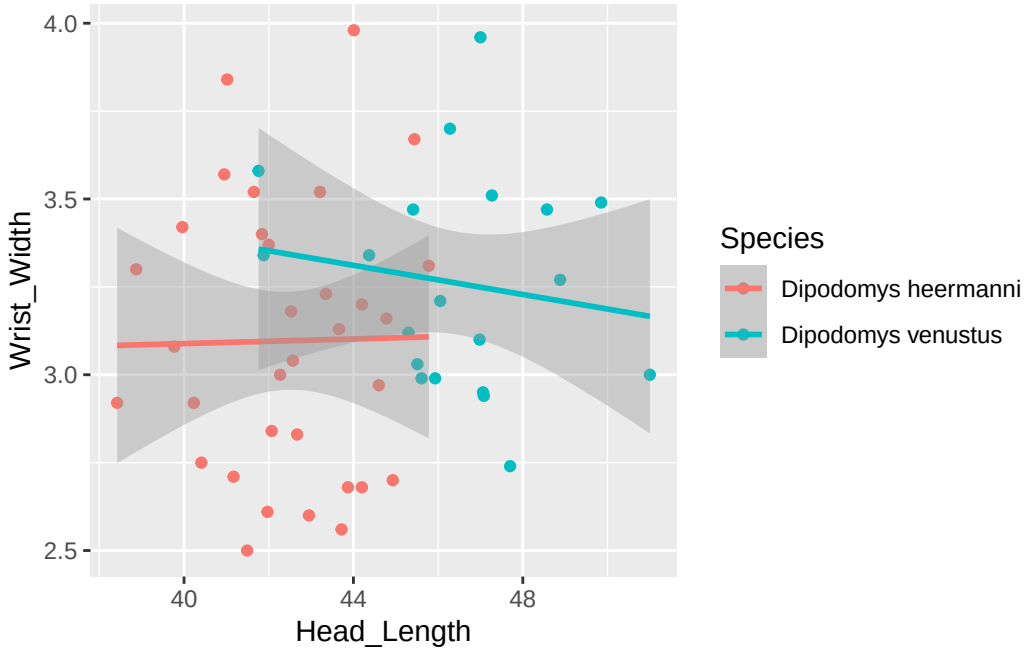
```r
# Also remove wrist width values less than 2.47 mm
df.o$Wrist_Width[df$Wrist_Width < 2.47] <- NA

#Regraph with the new outlier removals
df.o |>
  ggplot(aes(y= Wrist_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 7 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

The wrist widths seem to be very variable and not too different between the species. If there is a difference it is very slight. May be inclined not to include this variable in the PCA.

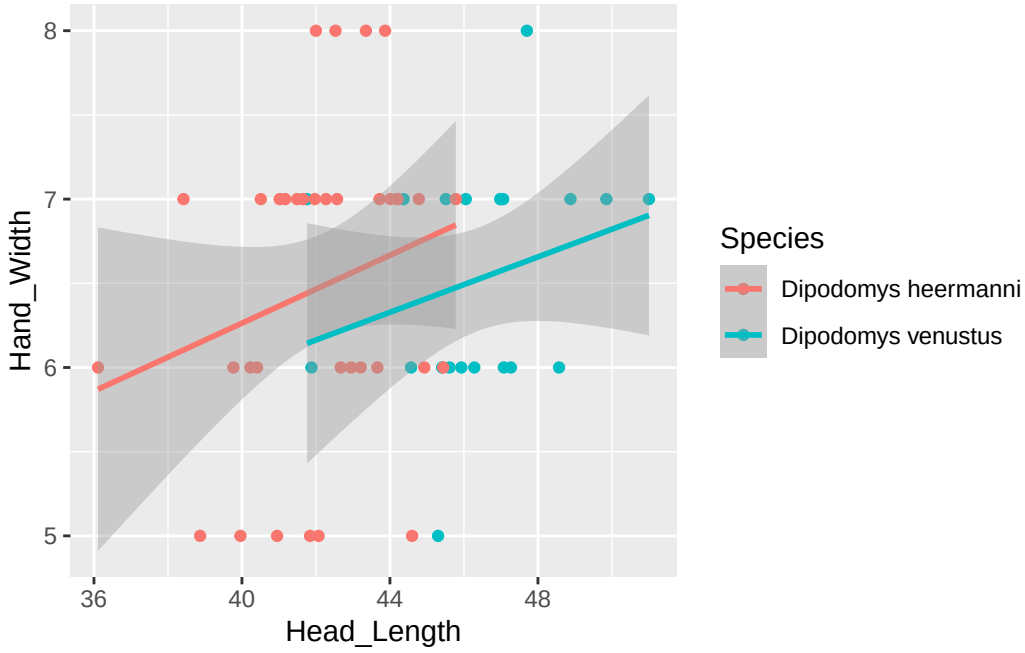
12) Hand width

```
#Graph hand width versus head length
df |>
ggplot(aes(y= Hand_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 4 rows containing non-finite outside the scale range
(`stat_smooth()`).

Warning: Removed 4 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

There don't seem to be too many true outliers in this. It is a little weird since it is such a small measurement that was measured with the ruler not the caliper meaning some precision was lost.

Interpretation:

Head length seems to be doing most of the differentiating in this measurement. It is likely that the hand widths do not vary greatly between these species.

13) Head length

This is our size proxy measurement.

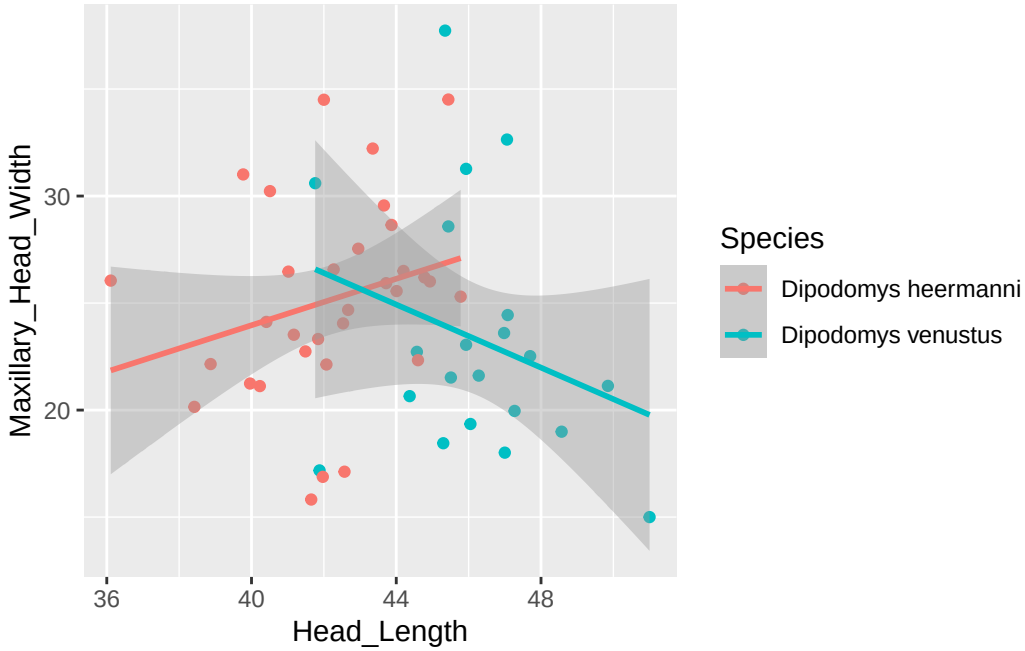
14) Maxillary head width

```
#graph maxillary head width vs head length
df |>
  ggplot(aes(y= Maxillary_Head_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 7 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 7 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

The venustus with a maxillary head width greater than 36 seems like an outlier

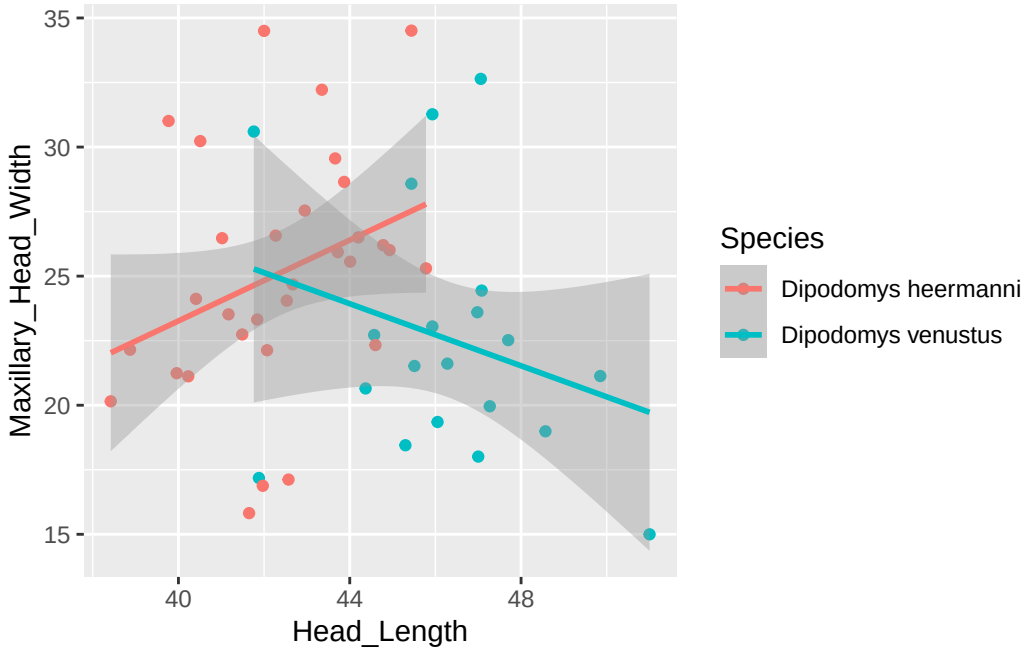
```
#Remove outlier greater than 36 mm maxillary head width
df.o$Maxillary_Head_Width[df$Maxillary_Head_Width > 36] <- NA

#Regraph data without the outlier
df.o |>
  ggplot(aes(y= Maxillary_Head_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 9 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 9 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Interpretation:

This data is pretty spread and there seems to be consistent crazy outliers for both species. There does seem to be clustering for both, however that would mean when taking into account both head width and length one might be able to more accurately identify species.

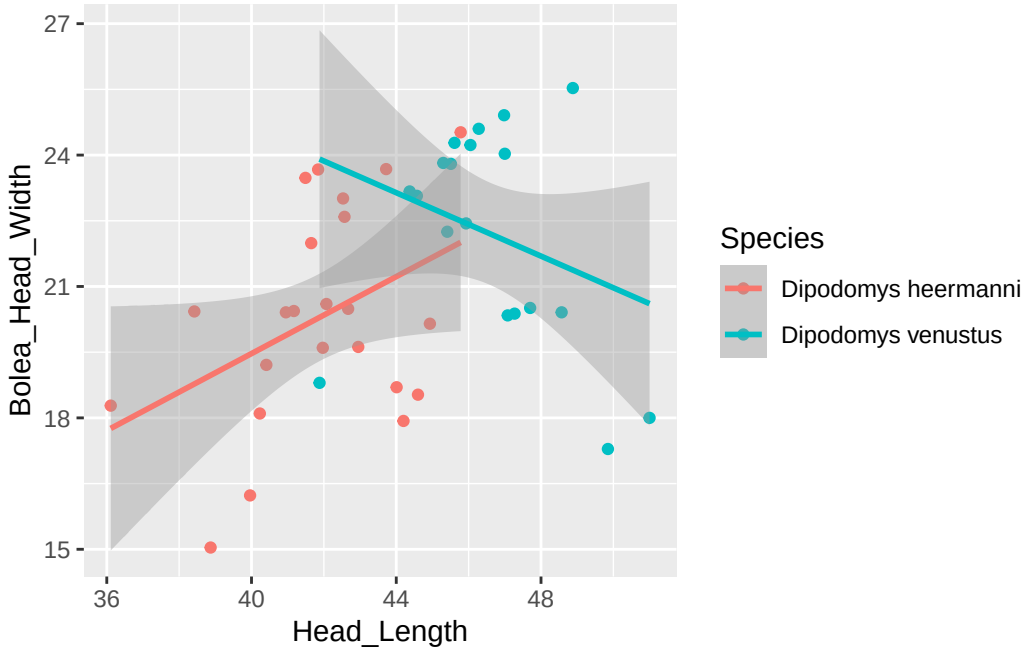
15) Bolea head width

```
#Graph the bolea head width vs head length
df |>
  ggplot(aes(y= Bolea_Head_Width, x= Head_Length, color =  Species))+
  geom_point()+
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 18 rows containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 18 rows containing missing values or values outside the scale range
(`geom_point()`).
```



Outliers:

There doesnt seem to be too many really crazy outliers, except the venustus in the middle of the heermanni. for that individual it isnt that its head length or bolea head width measurement is out of the ordinary for its species, just the combination of both values together that makes it stand out.

Interpretation:

It is interesting to see different slopes for these species. This data suggests that as the skull of heermanni grow the bolea grow wider and as the skull of venustus grow the bolea grow narrower. There is documentation in the literature of heermanni being known as a wider faced group and the venustus as the narrowed-faced group. Is this the same as what is documented there or does this data seem weird for that?